ITDS231
Lecture 6.1

Data Link Layer
Media Access Control Protocols

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Some slides from
- ITCS323/2022s2
- NETACAD Network Essentials/ITN7.02

มหาวิทยาลัยมหิดล
Mahidol University

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วนโดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 10 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

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Review: Physical and Logical Topologies

[CISCO NetAcad ITN7.0]

The topology of a network is the arrangement and relationship of the network devices and the interconnections between them.

There are two types of topologies used when describing networks:

- **Physical topology** – shows physical connections and how devices are interconnected.
- **Logical topology** – identifies the virtual connections between devices using device interfaces and IP addressing schemes.

WAN Topologies

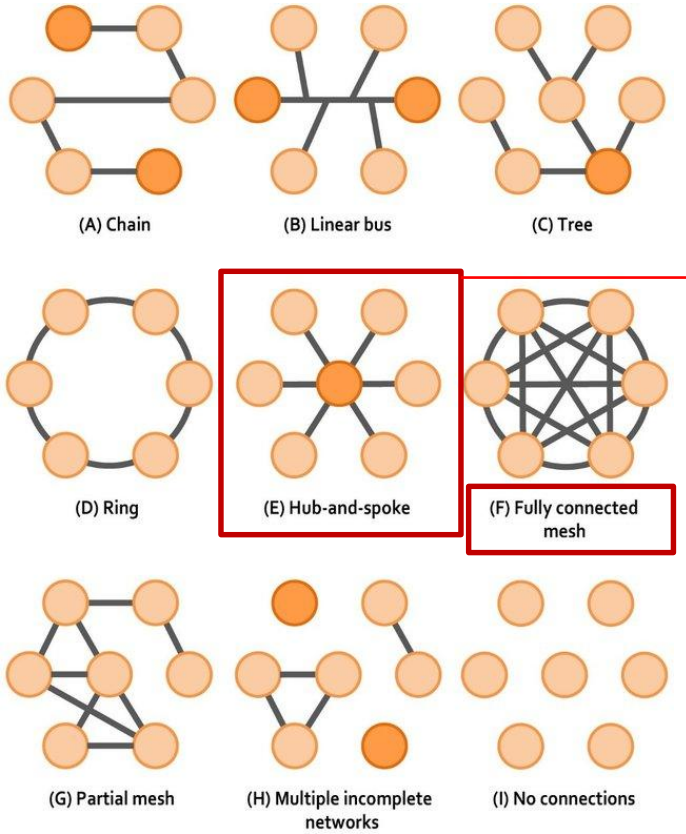
[CISCO NetAcad ITN7.0]

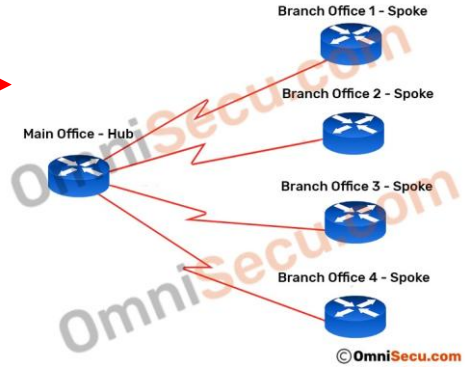
There are three common physical WAN topologies:

- **Point-to-point** – the simplest and most common WAN topology. Consists of a permanent link between two endpoints.
- **Hub and spoke** – similar to a star topology where a central site interconnects branch sites through point-to-point links.
- **Mesh** – provides high availability but requires every end system to be connected to every other end system.

https://networklessons.com/cisco/cisco-sd-wan/cisco-sd-wan-hub-and-spoke-topology

https://www.youtube.com/watch?v=w6ofZqPNbCk



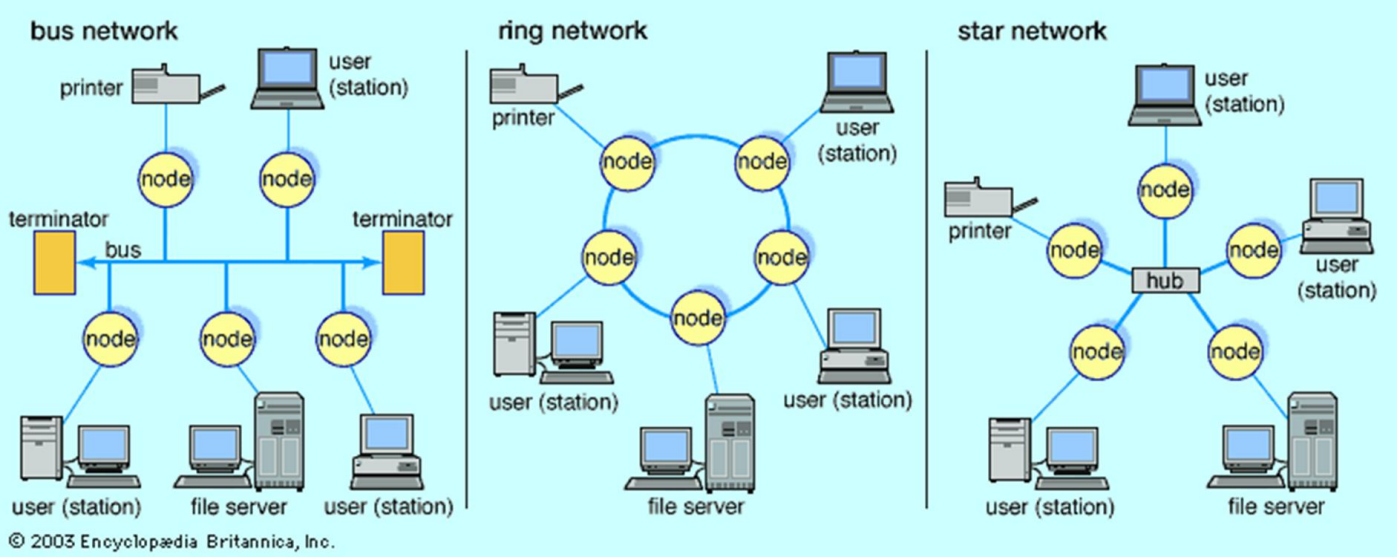


LAN Topologies

- **Star or Extended Star**
  - End devices on LANs are typically interconnected using Star (via Hub/Switch)
  - Each device is easy to install, very scalable and easy to troubleshoot.
- **Early Ethernet and Legacy Token Ring** technologies provide **two** additional topologies:

✓ **Bus** – All end systems chained together and terminated on each end.

✓ **Ring** – Each end system is connected to its respective neighbors to form a ring.



Transmission Mode

Simplex/Half-Duplex/Full-Duplex Communication

Simplex communication

- One way communication or unidirectional form of communication.
- Only device is dedicated as a receiver (e.g. TV, radio) or otherwise a transmitter (e.g. remote control).

Half-duplex communication

- Only allows **one device to send or receive at a time** on a shared medium.
- Used on WLANs and legacy bus topologies with Ethernet hubs.

Full-duplex communication

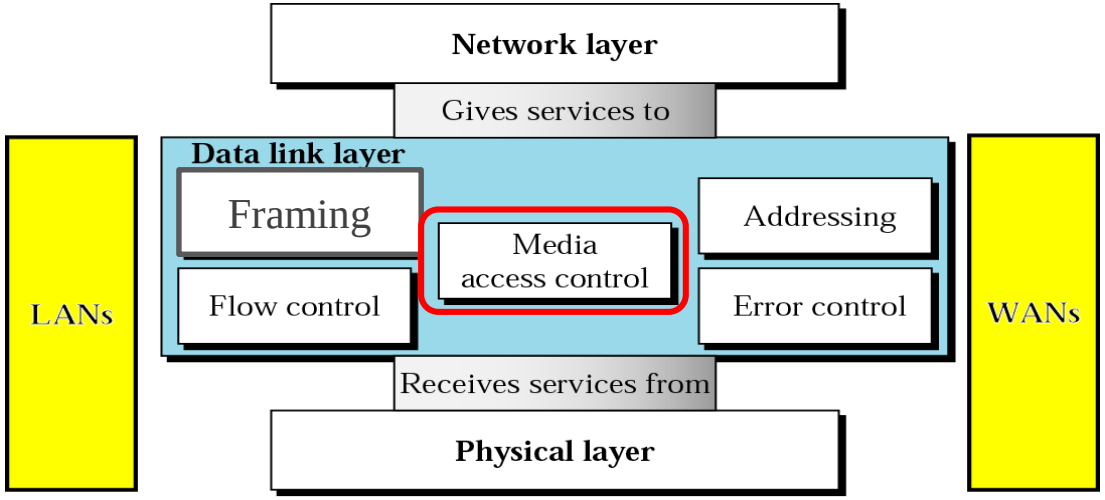
- Allows **both devices** to **simultaneously** transmit and receive on a shared medium.
- Ethernet switches operate in full-duplex mode.

https://www.youtube.com/watch?v=RzAXEQ4XsSI

Multiple Access Control (1/2)

- Multiple Access Control refers to a set of procedures used to restrict the transmission of data that the senders can send **in a shared channel or broadcast link** so that the data collision is minimized.

- Multiple Access Protocols regulate or control access to the link.



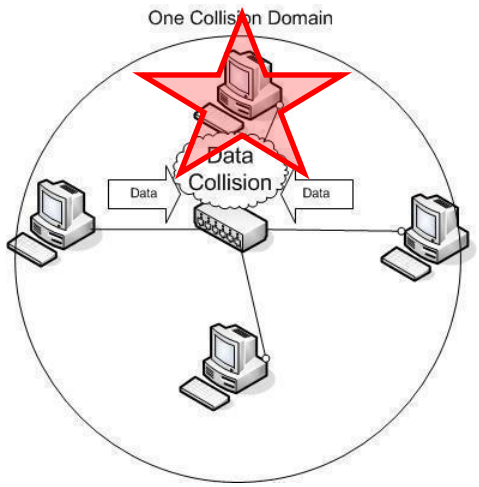
Multiple Access Control (2/2)

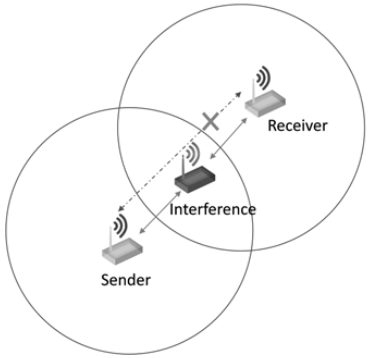
- Traditionally in LAN, all stations are connected to a shared transmission medium (bus topology or star topology using hub).
- Simultaneous attempts to access the medium by more than one station happen with high probability.
- When two or more stations send data at the same time, a collision is the result.
- To prevent this situation, each protocol in the data link layer must have a Medium Access Control (MAC) method to ensure that there is no conflict among the stations.

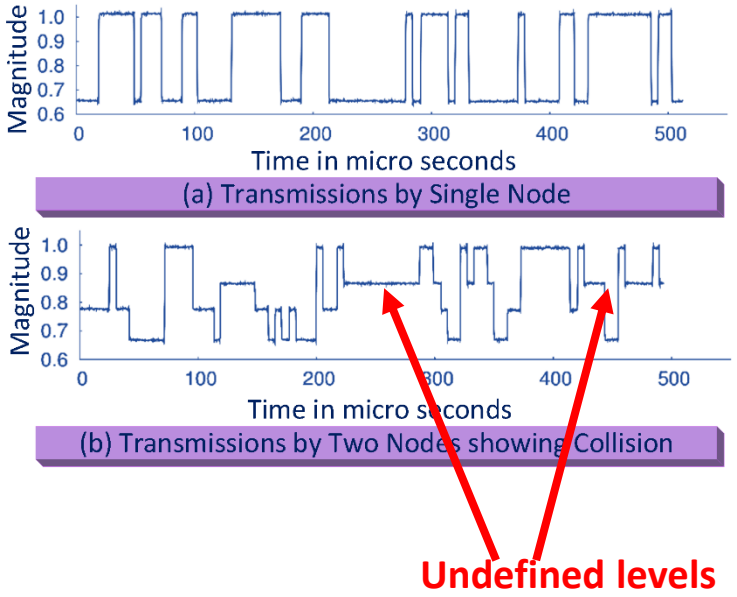
Note: MAC (Media Access Control) Address

Collision

- If two or more computers transmit at the same time, a collision can occur.
- Collision can happen in the air and may not be in the same network as long as the signals are at the same frequencies.





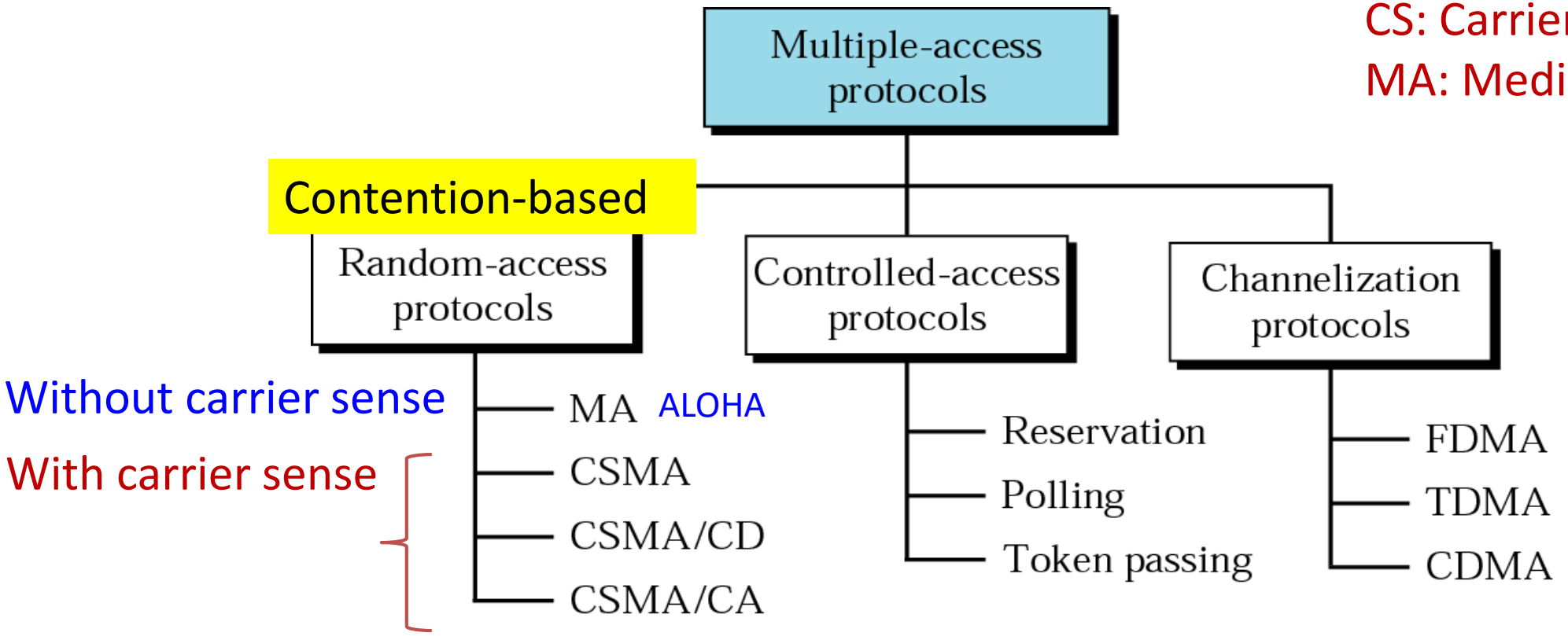


Multiple-Access Control Methods (1/2)

Objectives

- How to prevent signals from colliding with each other?
- We have to control how and when each sender speaks.

CS: Carrier Sense
MA: Media Access



Multiple Access Control Methods (2/2)

- **Contention-based access**
  - Performance:
    - Best for bursty type traffic, and light-to-moderate traffic loads
    - (+) Simple to implement
    - (-) Performance is poor for heavy traffic
  - All nodes operating in half-duplex, competing for use of the medium. Examples are:
    - ✓ Carrier sense multiple access with collision detection (CSMA/CD) as used on legacy bus-topology Ethernet.
    - ✓ Carrier sense multiple access with ALOHA, collision avoidance (CSMA/CA) as used on Wireless LANs.
- **Controlled access**
  - Deterministic access where each node has its own time on the medium.
  - Used on legacy networks such as Token Ring and ARCNET.
  - ✓ **Round Robin-based:** Polling (Centralized), Token Passing (Decentralized)
  - ✓ **Reservation-based** : BRAP - Broadcast Recognition with Alternating Priorities (Centralized), DQDB (IEEE802.6), DAMA (Decentralized)

Multiple-Access Control Protocols

	Fast Ethernet	Gigabit Ethernet	Fibre Channel	Wireless LAN
Data Rate	100 Mbps	1 Gbps, 10 Gbps	100 Mbps–3.2 Gbps	1 Mbps–54 Mbps
Transmission Media	UTP, STP, optical Fiber	UTP, shielded cable, optical fiber	Optical fiber, coaxial cable, STP	2.4-GHz, 5-GHz microwave
Access Method	CSMA/CD	Switched	Switched	CSMA/Polling
Supporting Standard	IEEE 802.3	IEEE 802.3	Fibre Channel Association	IEEE 802.11

Working group:
https://en.wikipedia.org/wiki/IEEE_802
https://mentor.ieee.org/802/bp/StartPage

Contention Based

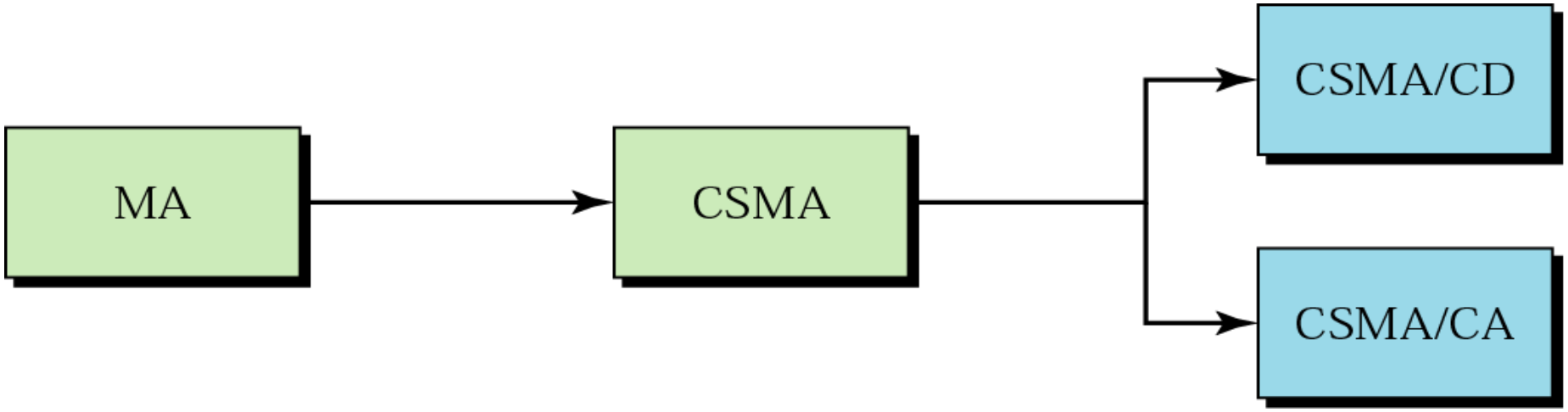
Contention Based: CSMA/CD (1/2)

- Used by legacy Ethernet LANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision detection process to govern when a device can send and what happens if multiple devices send at the same time.
- Random Access Method
  - ✓ each station has the right to the medium without being controlled by any other station
  - ✓ If more than one station tries to send, there is an access conflict == collision! And the frame will be either destroyed or modified.
  - ✓ No scheduler, No rules → contention (competition with other stations)
  - ✓ If access conflict occurs, then there is collision

Contention Based: CSMA/CD (2/2)

CSMA/CD collision detection process:

- Devices transmitting simultaneously will result in a signal collision on the shared media.
- Devices detect the collision (transmission medium is busy).
- Devices wait a random period of time and retransmit data.



Contention Based: CSMA/CA

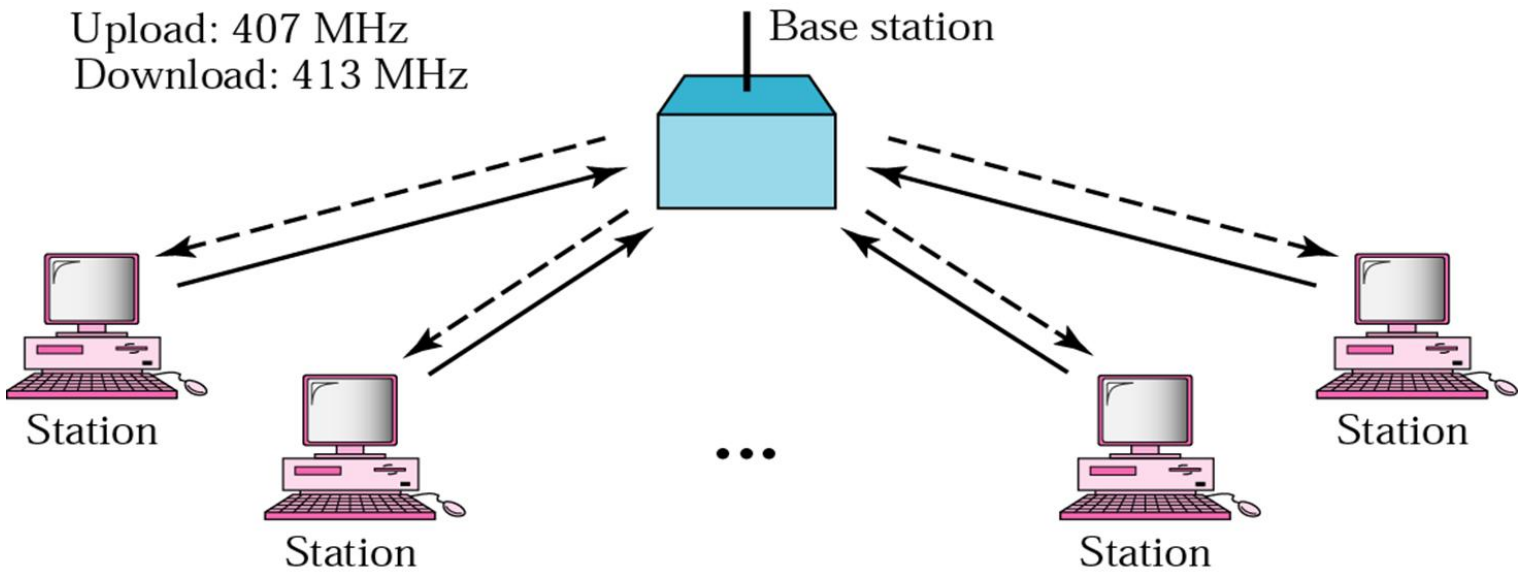
- Used by IEEE 802.11 WLANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision avoidance process to govern when a device can send and what happens if multiple devices send at the same time.

CSMA/CA collision avoidance process:

- When transmitting, devices also include the time duration needed for the transmission.
- Other devices on the shared medium receive the time duration information and know how long the medium will be unavailable.

Multiple Access (MA)

- ALOHA is the earliest random-access method, developed in the early 1970s
- Designed to be used on a radio (wireless) local area network (LAN) with a data rate of 9600 bps.



ALOHA Protocol

- Multiple Access: any station sends a frame when it has a frame to send

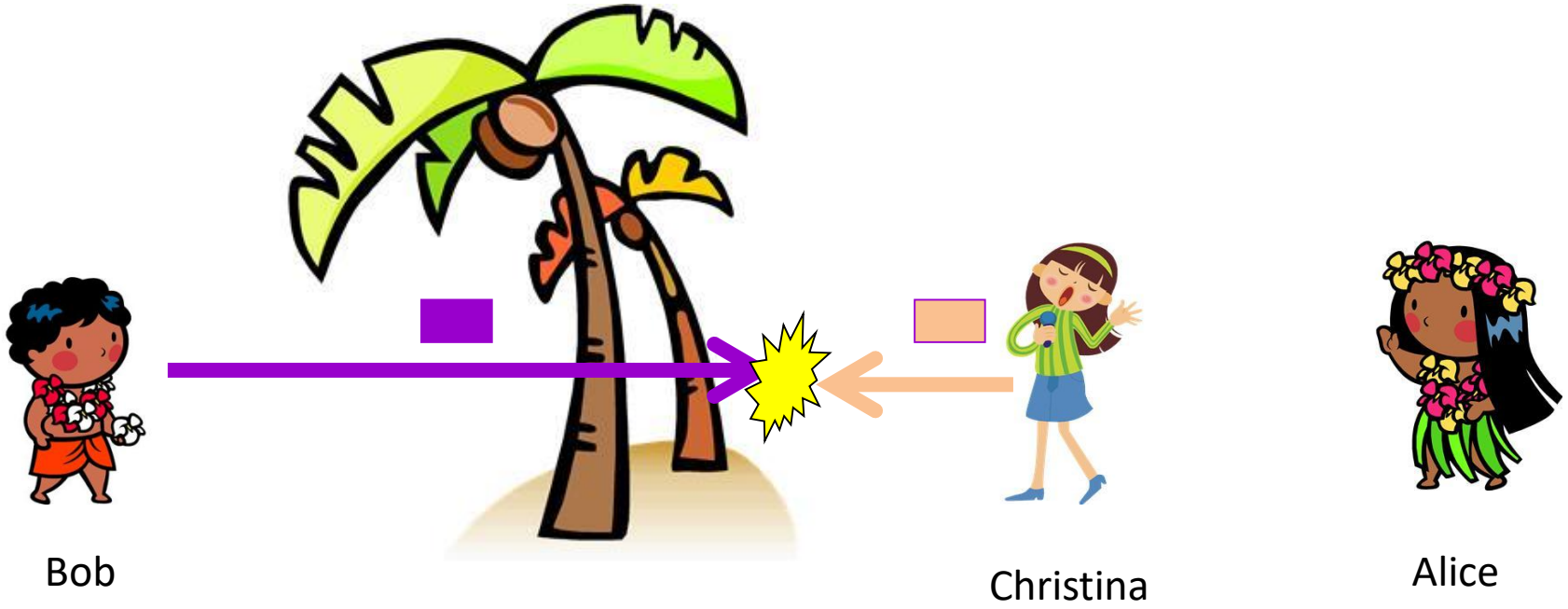
- No Carrier Sense: A station that has a frame to send does not check to see if any other station is sending (there is no sensing of the carrier frequency). The station does not check to see if the medium is free. For wireless medium, the stations can be so far from each other that a station cannot sense that another station is sending.

- No Checking for Collision: after sending, the station does not listen to medium to see if there is a collision.

- Acknowledgement: after ending, the station waits for an acknowledgement (explicit or implicit). If it does not receive the acknowledge during the allotted time, then retransmit.

Packet Collisions in the Wireless Network

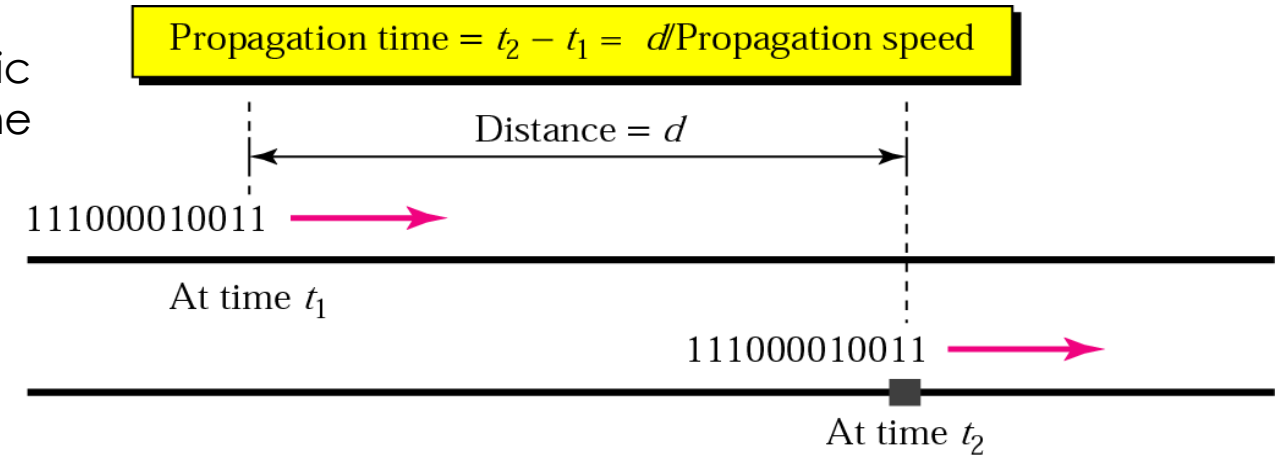
Bob wants to talk to Alice.
Bob does not know that his packet got a collision behind the coconuts tree!



Acknowledgement

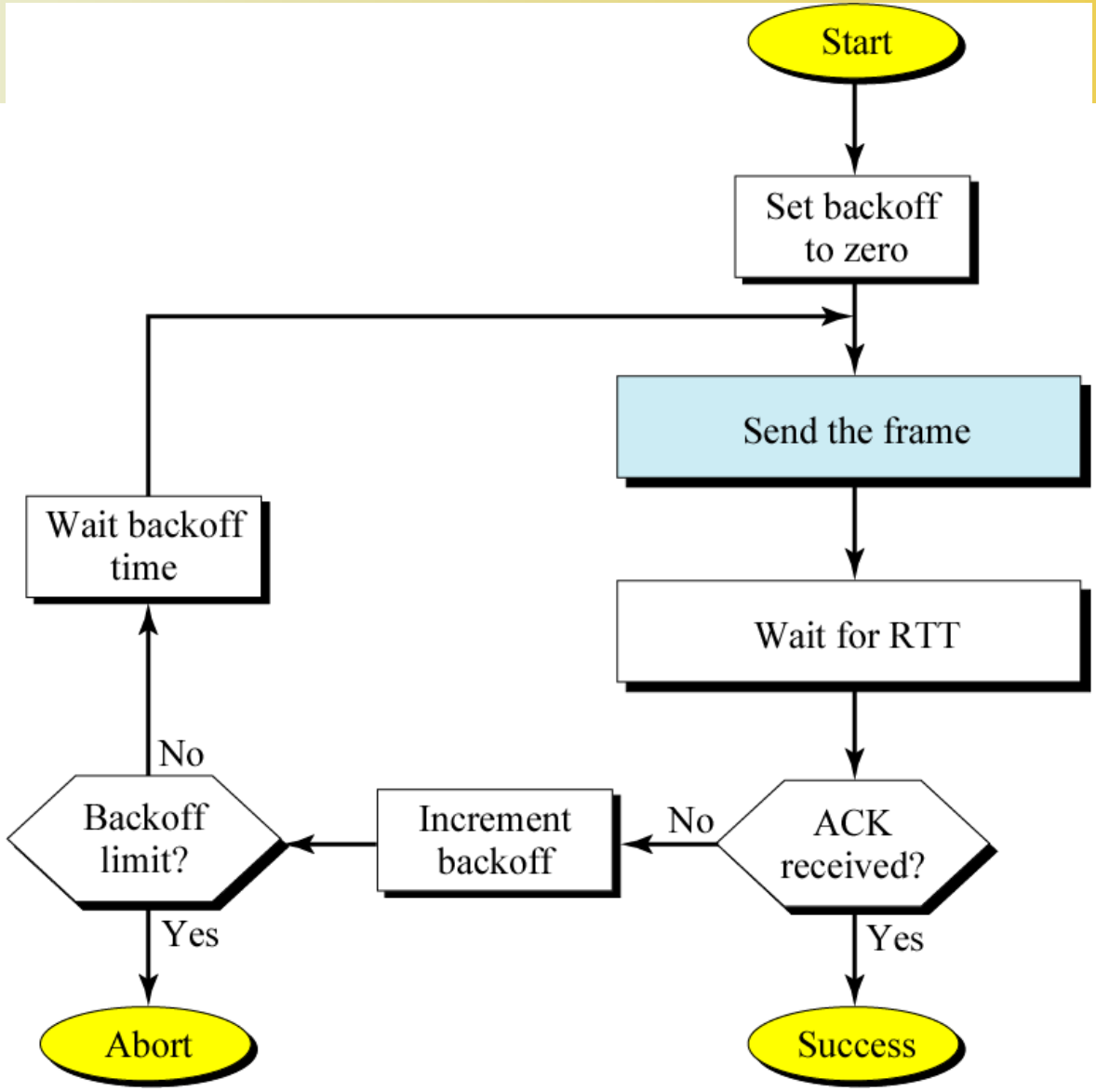
- Multiple Access
  - ✓ Any station sends a frame when it has a frame to send
- Acknowledgement
  - ✓ After sending the frame, the station waits for an acknowledgement
  - ✓ If not receive an ACK during the allotted time, which is 2 times the maximum propagation delay, it assumes that the frame is lost. Then, it tries to send again after a random amount of time

The propagation speed of electromagnetic signals depends on the medium and on the frequency of the signal.



Pure ALOHA Protocol

Backoff timer



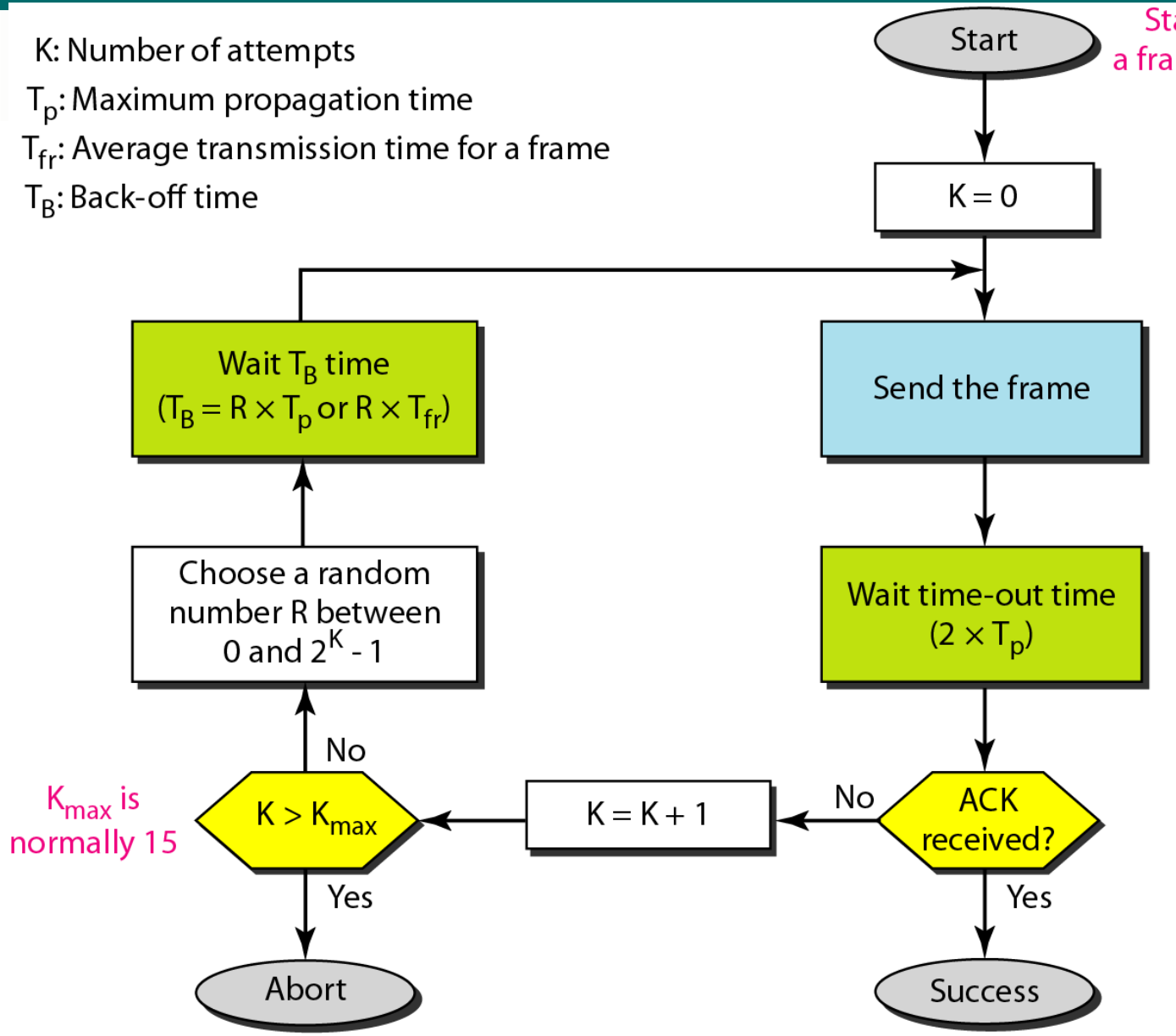
Pure ALOHA Protocol

K: Number of attempts

T_p: Maximum propagation time

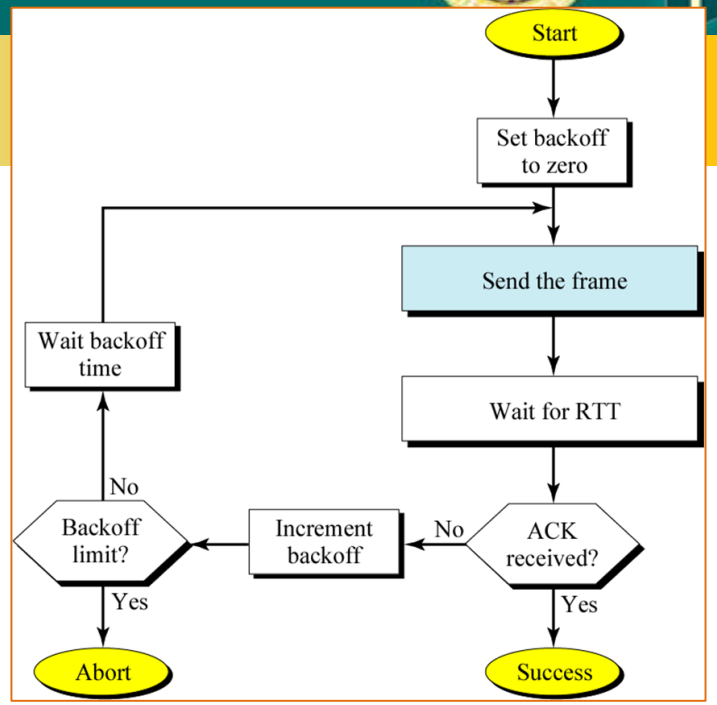
T_{fr}: Average transmission time for a frame

T_B: Back-off time



Station has a frame to send

K_{max} is normally 15



Example

Range (R) = 0 - 2^K -1

The stations on a wireless ALOHA network are a maximum of 600 km apart. If we assume that signals propagate at 3×10^8 m/s, we find Tp = (6×10^5) / (3×10^8) = 2 ms.
Now we can find the value of T_B for different values of K .

a. For K = 1, the range is {0, 1}. The station needs to generate a random number with a value of 0 or 1. This means that T_B is either 0 ms (0 × 2) or 2 ms (1 × 2), based on the outcome of the random variable.

b. For K = 2, the range (R) is {0, 1, 2, 3}. This means that T_B can be 0, 2, 4, or 6 ms, based on the outcome of the random variable, where Tp = 2ms.

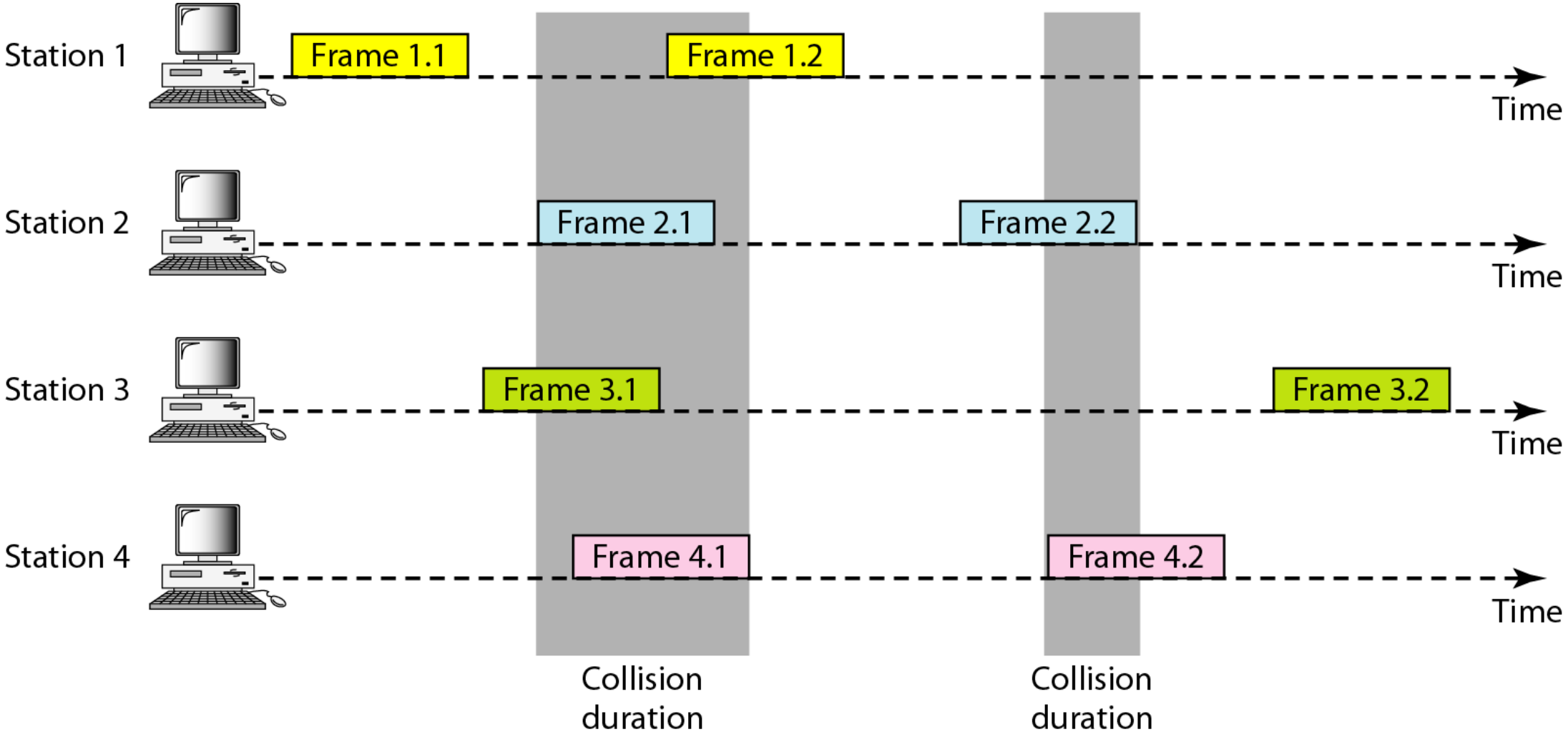
c. For K = 3, the range is {0, 1, 2, 3, 4, 5, 6, 7}. This means that T_B can be 0, 2, 4, . . . , 14 ms, based on the outcome of the random variable, where Tp = 2ms.

d. We need to mention that if K > 10, it is normally set to 10.

Backoff Time (T_B)= R x Tp

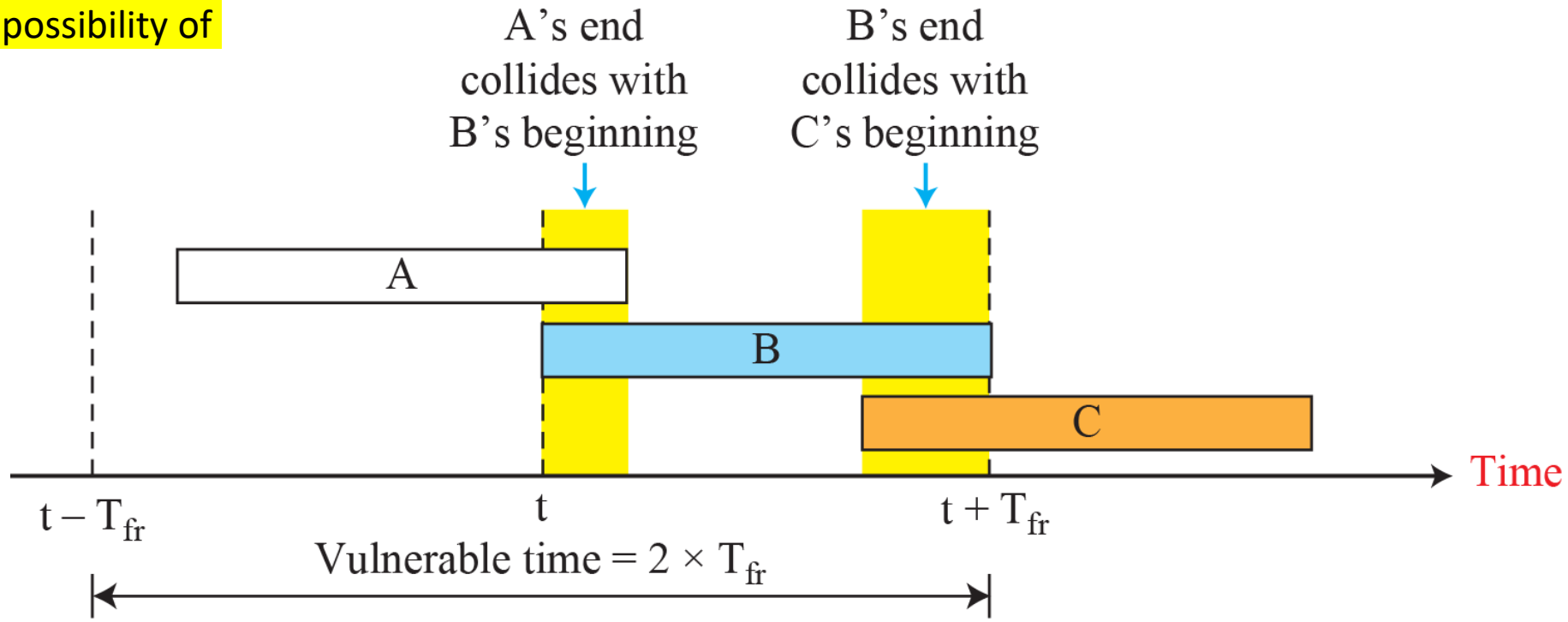
Pure ALOHA Protocol

Frames in a Pure ALOHA network

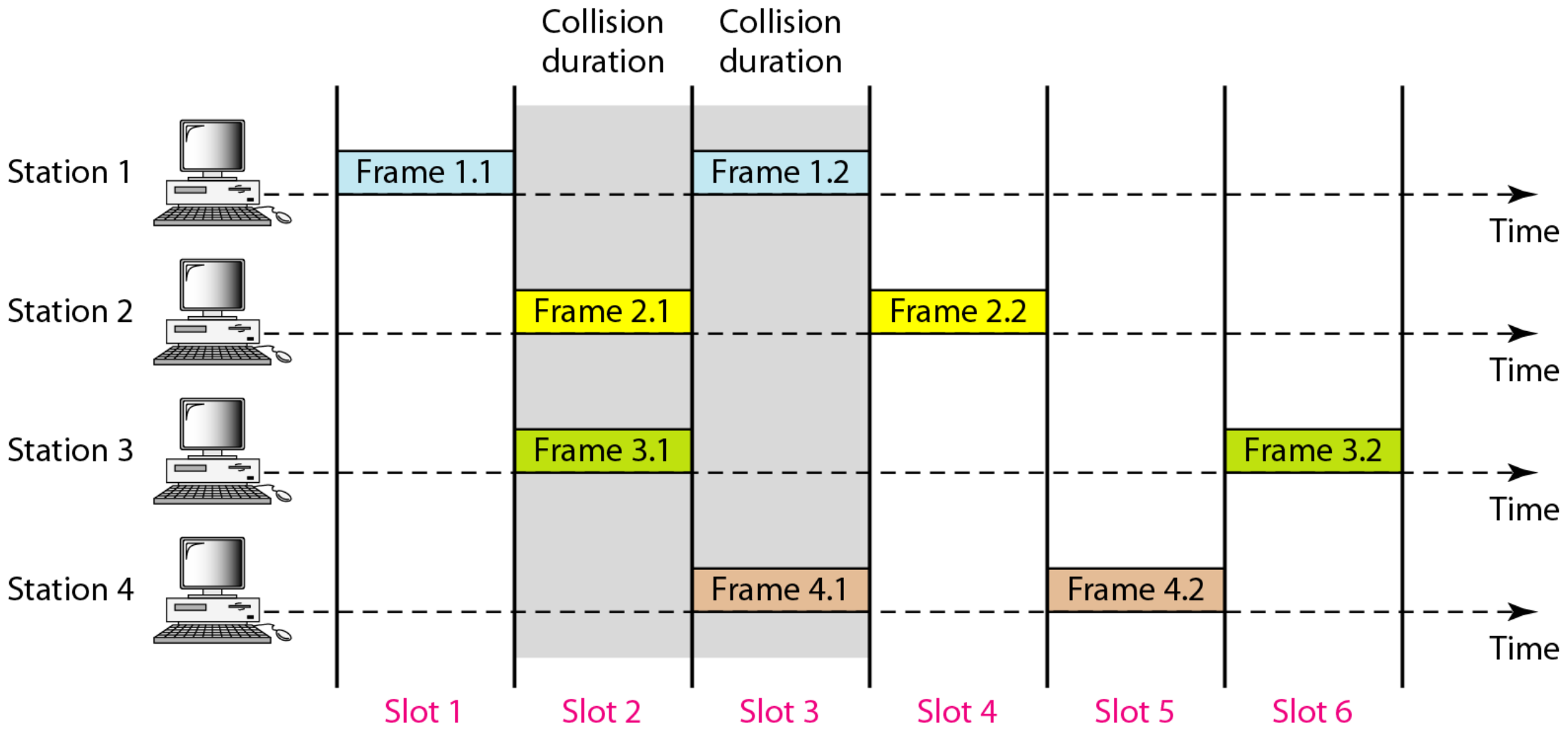


Vulnerable Time for a Pure ALOHA network

Vulnerable time is the length of time in which there is a possibility of collision.



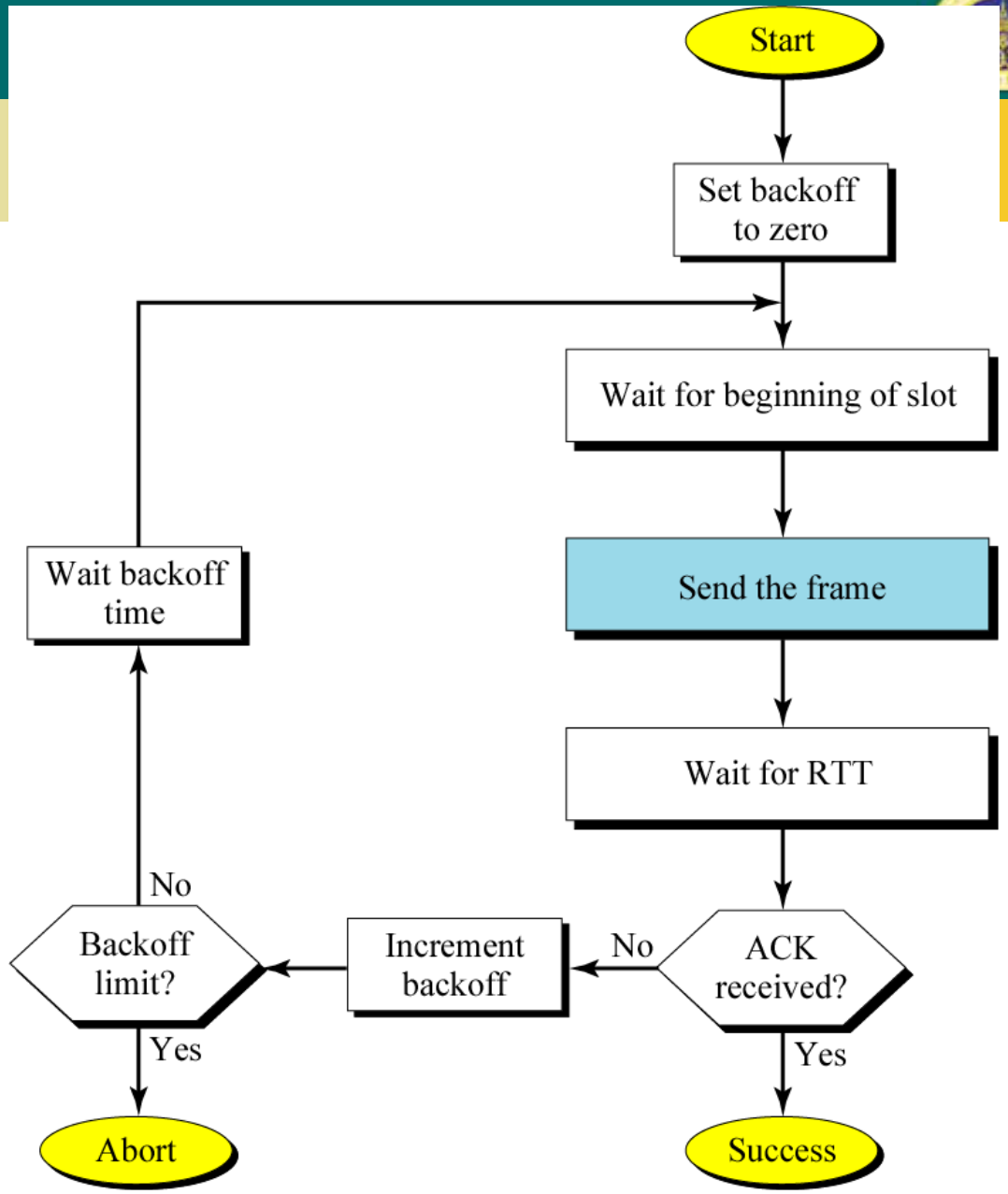
Frames in a Slotted ALOHA network



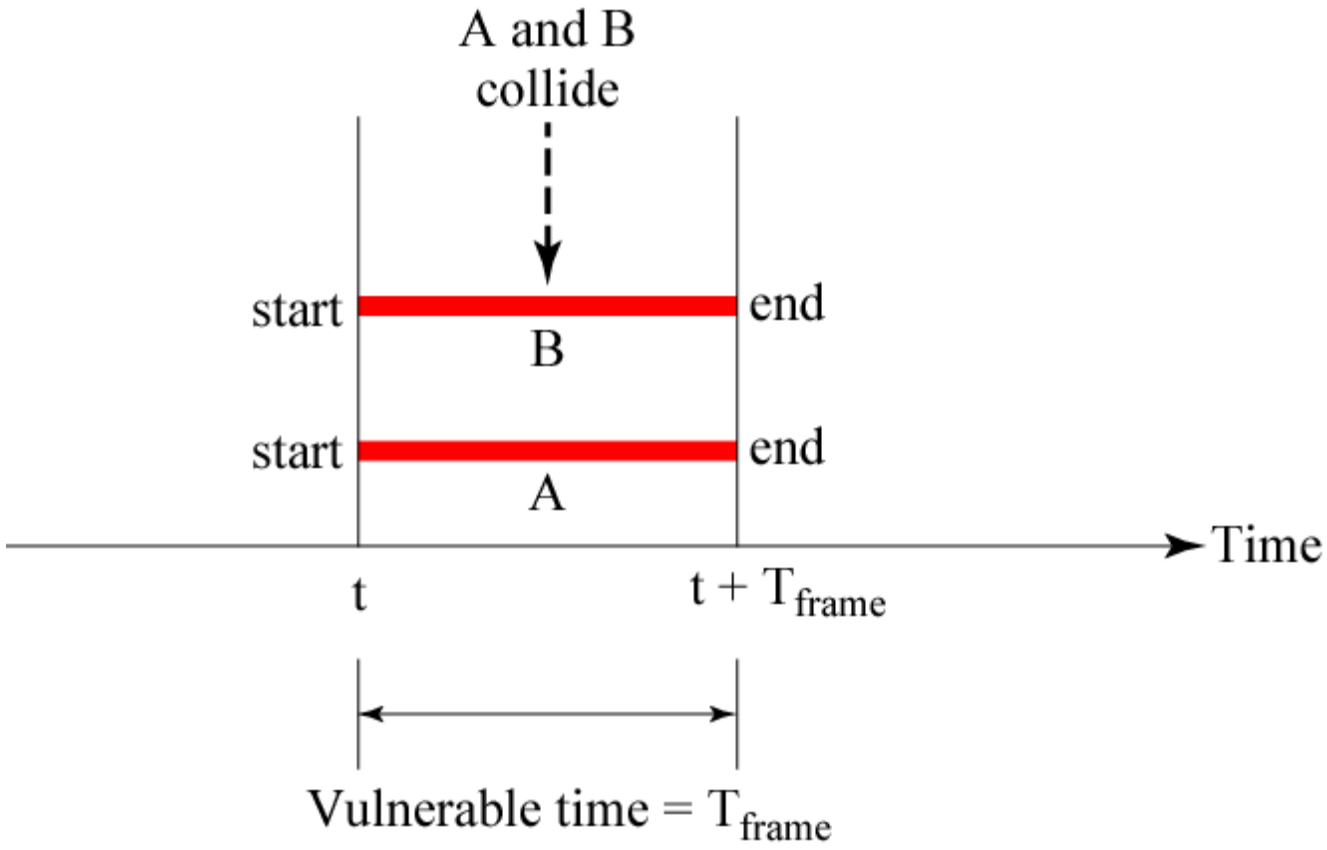
Slotted ALOHA Protocol

Frames in a Slotted ALOHA network

We divide the time into slots of T_{fr} and force the station to send only at the beginning of the time slot.



Vulnerable Time for a Slotted ALOHA network

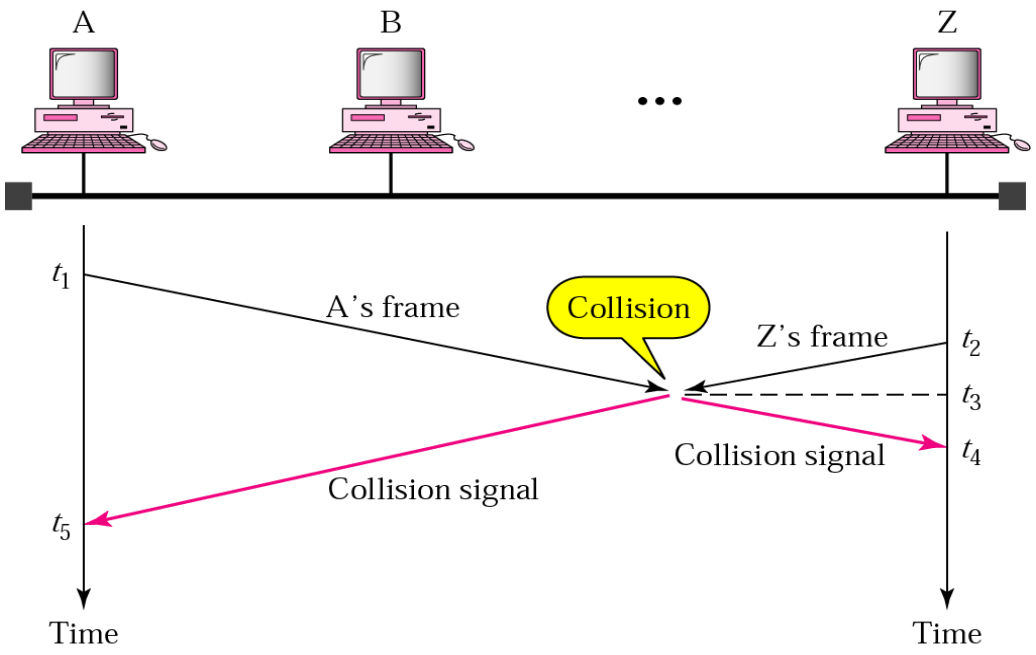


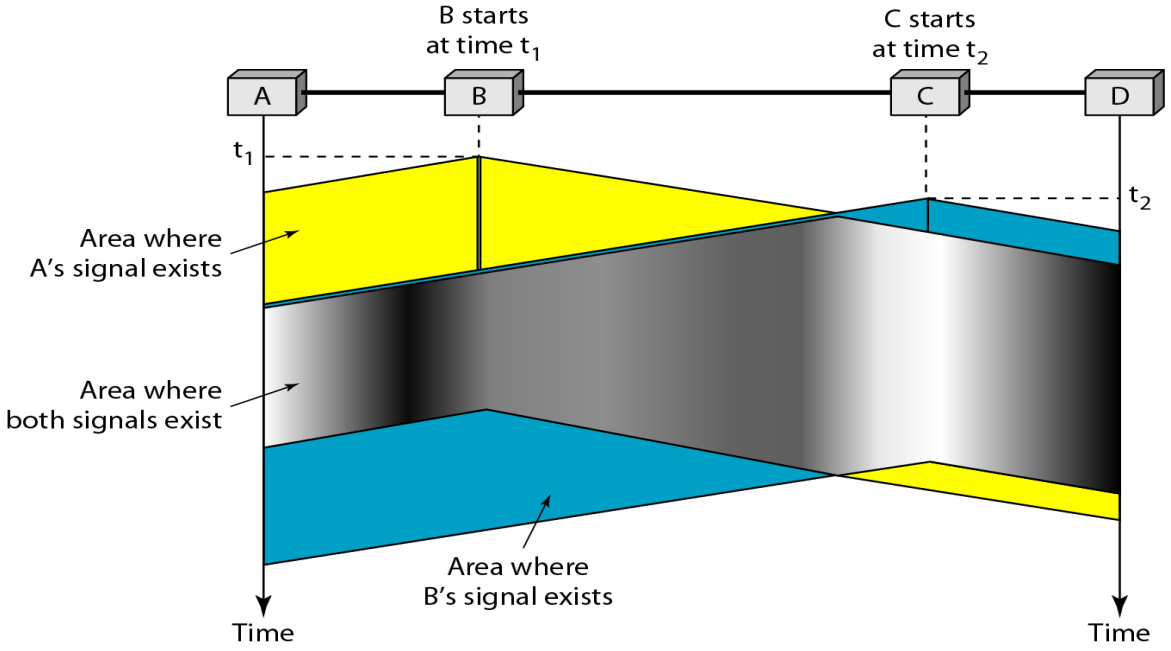
Carrier Sense Multiple Access (CSMA)

- To minimize the chance of collision, a station should sense the medium before trying transmission
  - ✓ *Sense before transmit*

- CSMA can reduce the possibility of collision, but it cannot eliminate it
  - ✓ Because of the propagation delay
  - ✓ The station may sense the medium and find it idle, only because propagation by another station has not yet reached the station

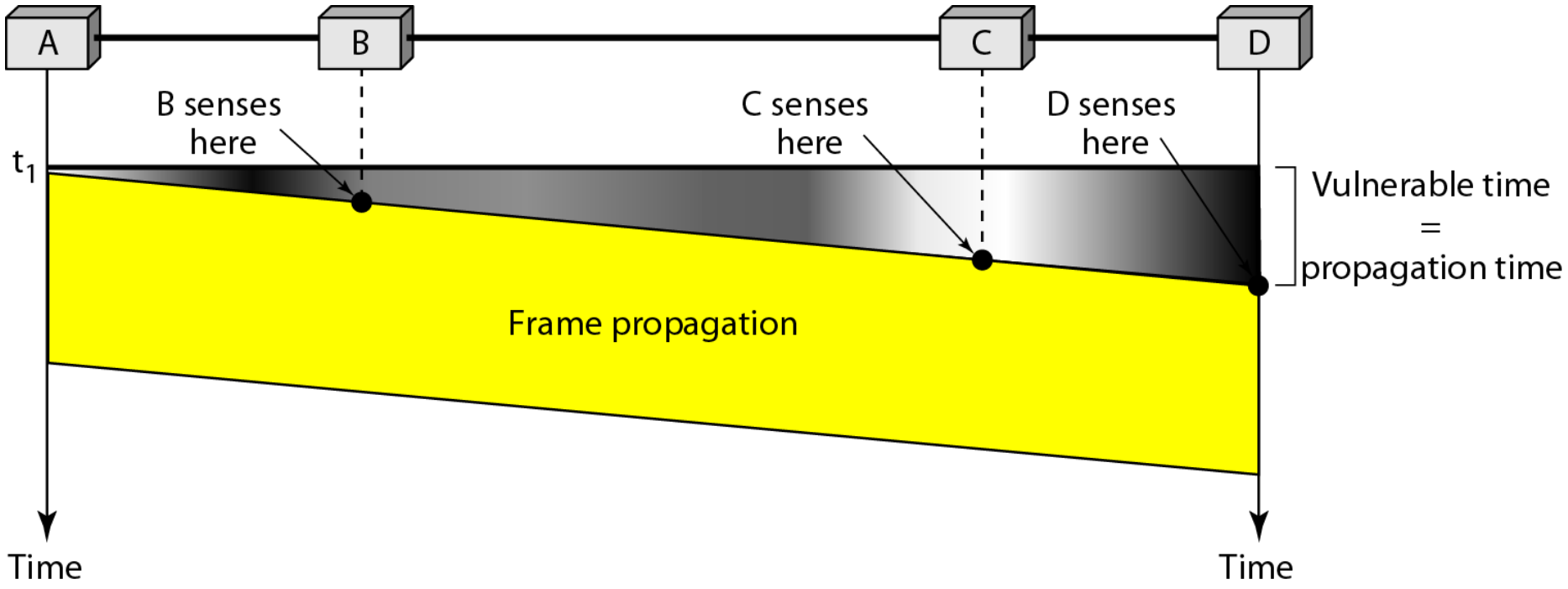
Collision in CSMA





Space/Time Model

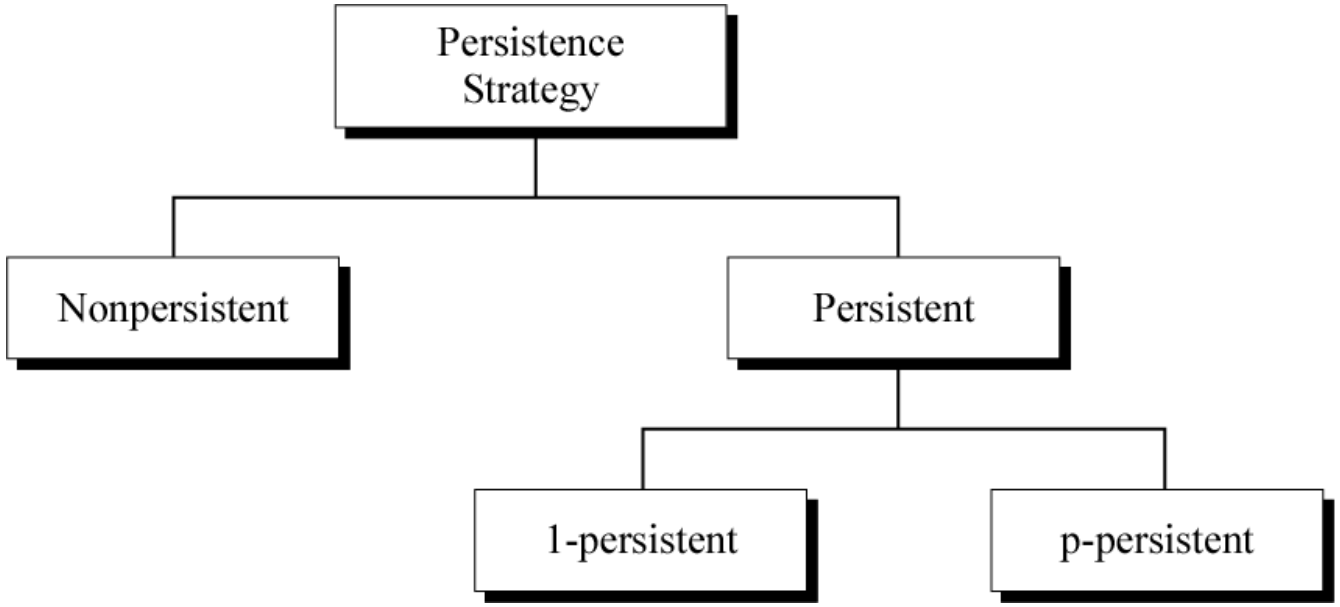
Vulnerable Time for a CSMA network



Persistence Strategy

Persistent strategies define what a station should do if, when sensing the medium, the station finds the channel busy.

- ✓ Nonpersistent
- ✓ Persistent
  - 1-persistent
  - *p*-persistent
    - – Probability $p = 0.2$
      - » Randomly generate number between 1 to 100
      - » If less than 20, the station will send

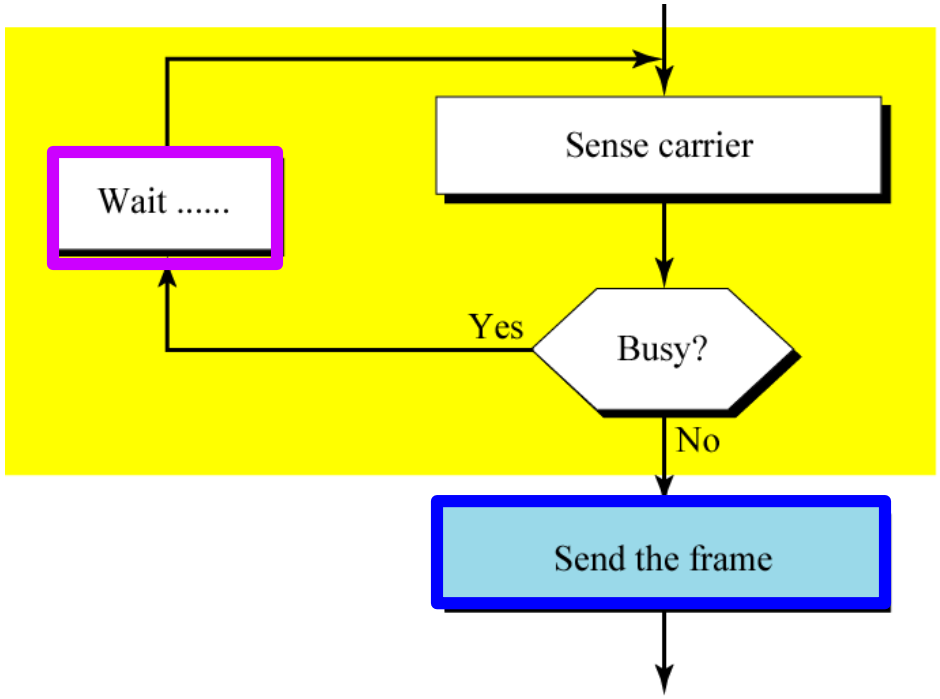


Non-Persistent Method vs 1-Persistent Method

Non-Persistent

If the channel is idle, the station sends immediately.

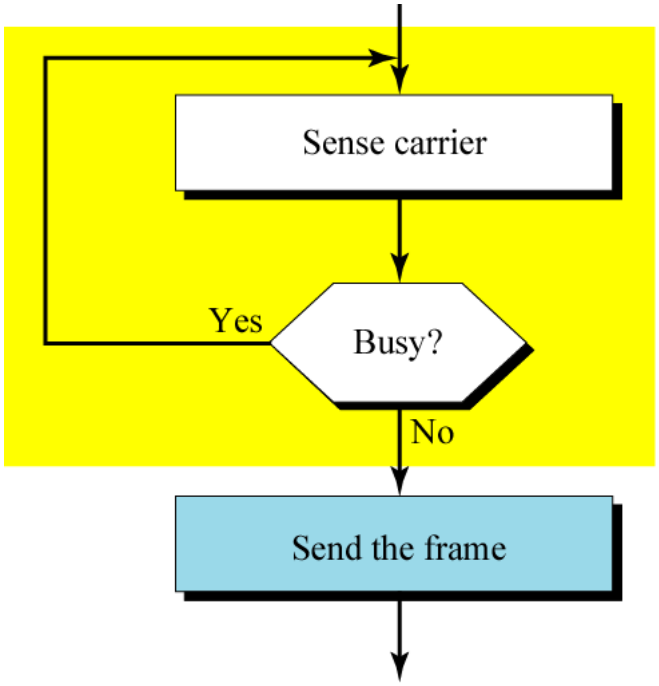
If not, wait for a random time and then senses the line again.



Nonpersistent strategy

1-Persistent

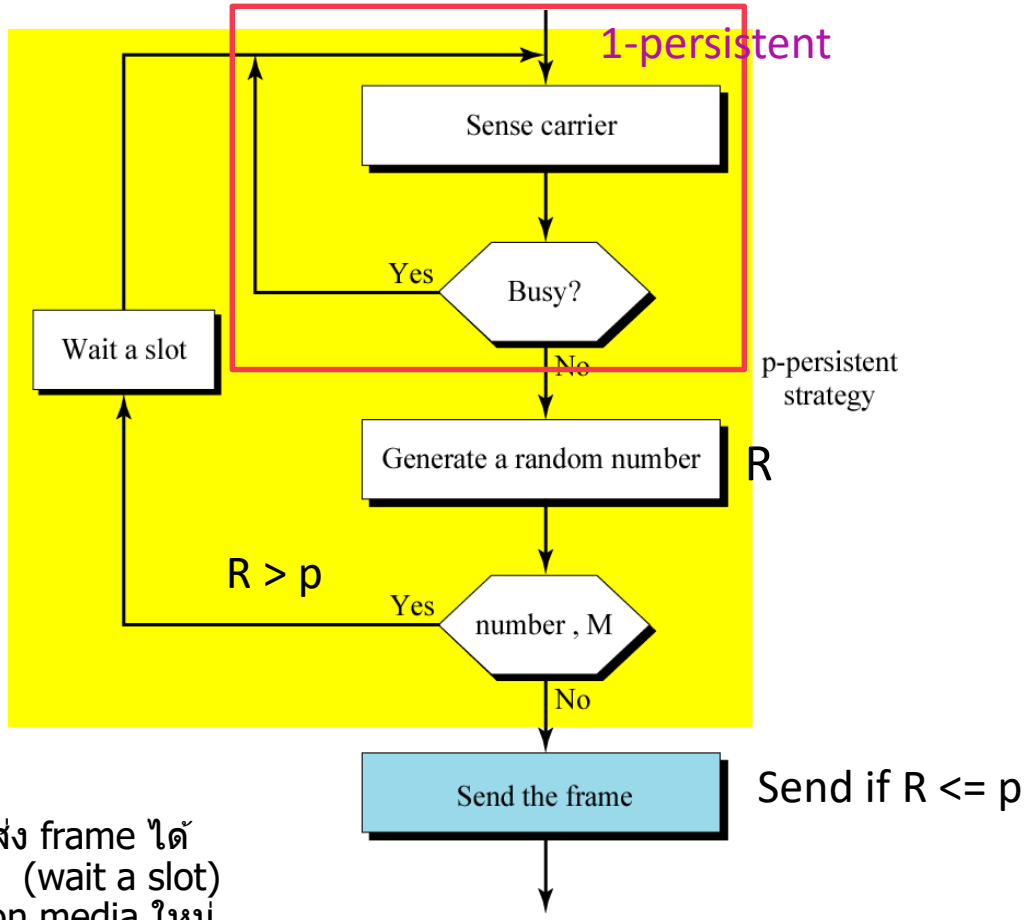
Similar to non-persistent but the station does not wait to sense the channel again when it finds the channel busy.



1-persistent strategy

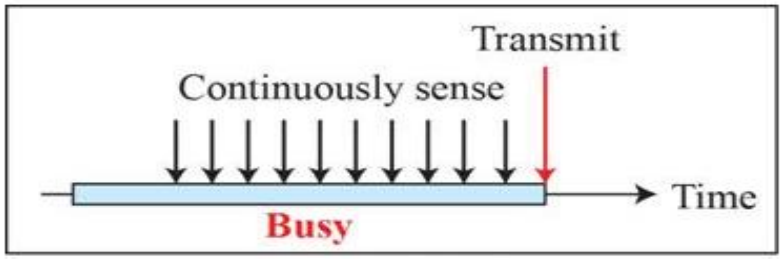
P-Persistent Method

Similar to 1-persistent but when the station finds the channel not busy, it transmits with probability p.

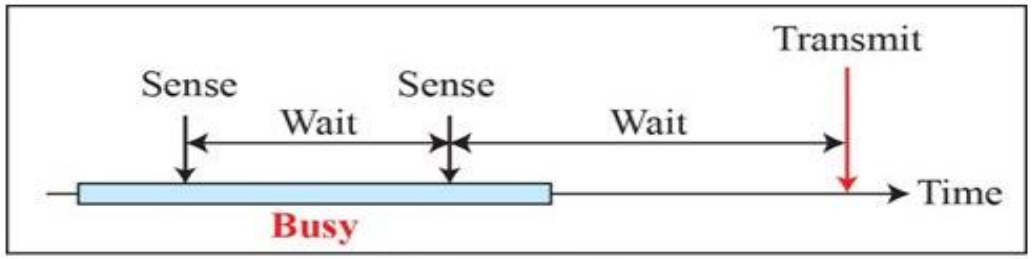


เช่น ถ้ากำหนดให้ p = 0.2/1 หรือ 20/100
- ถ้าค่า Random R = 0.1/1 หรือ 10/100 สามารถส่ง frame ได้
- ถ้าค่า Random R = 0.3/1 หรือ 30/100 ต้องคอย (wait a slot) เพื่อ sense carrier ด้วยการตรวจสอบ transmission media ใหม่

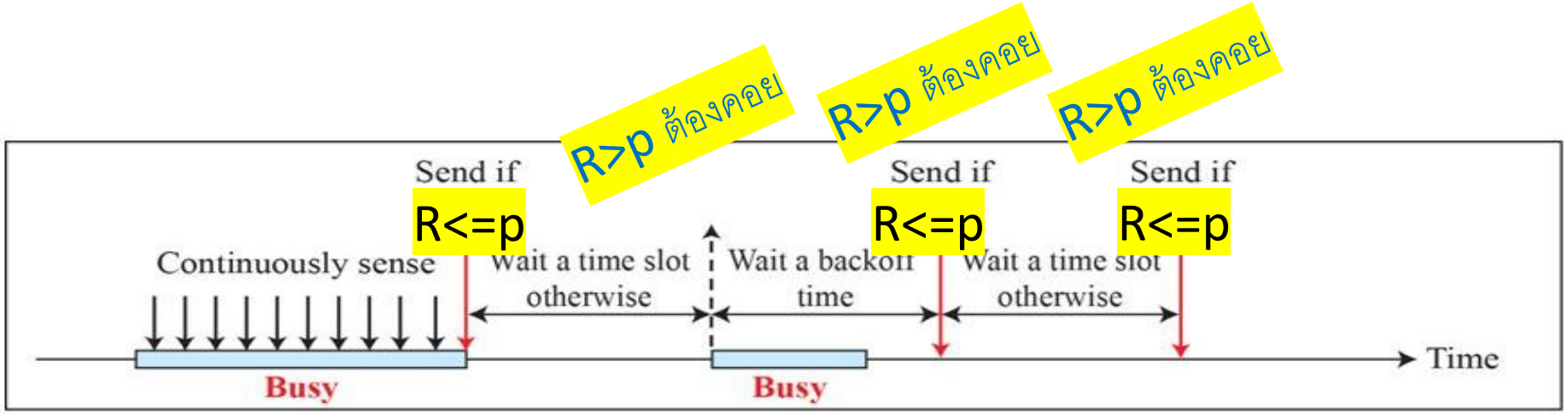
Persistent Strategy



a. 1-persistent



b. Nonpersistent



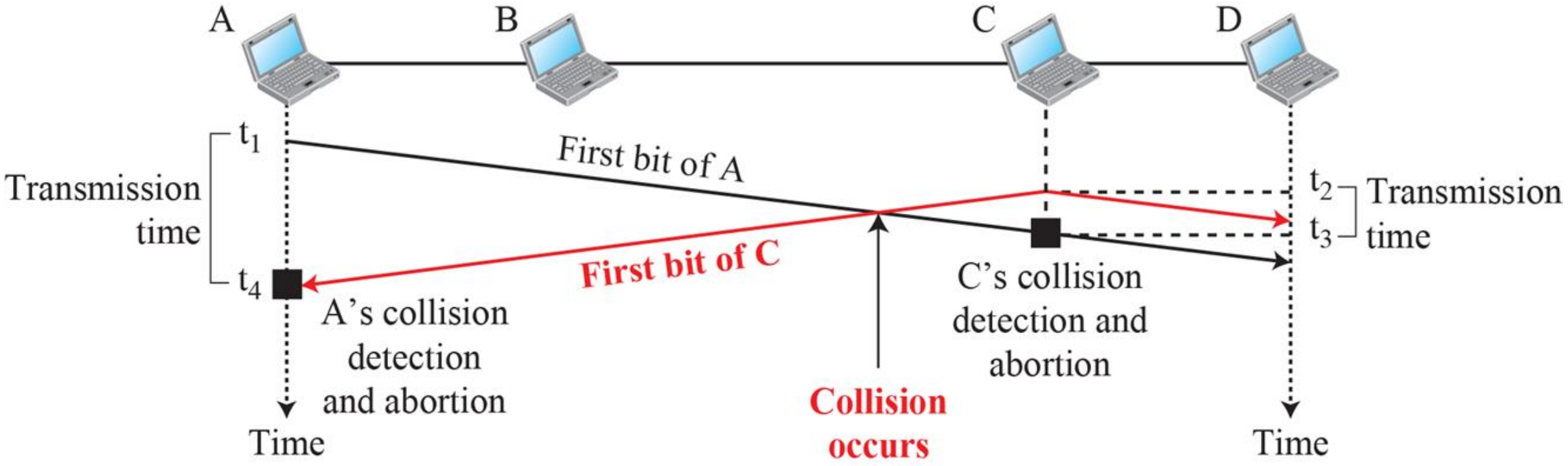
c. *p*-persistent

CSMA/CD

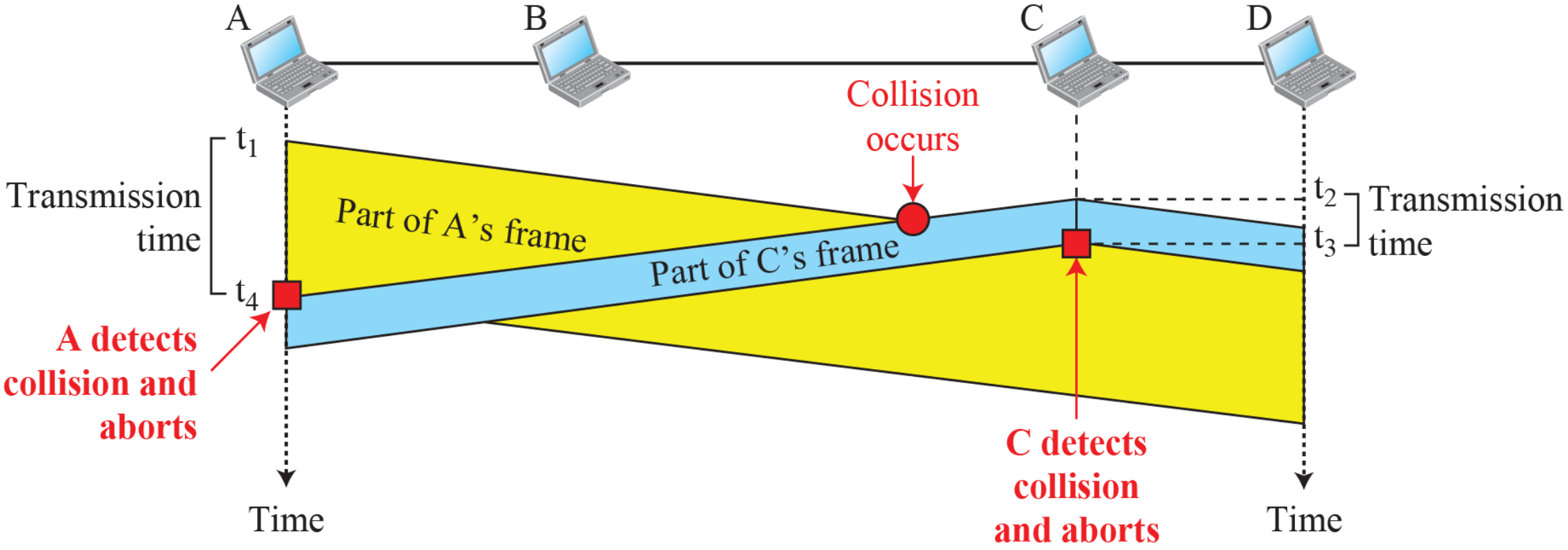
- CSMA is never implemented, because it does not define the procedure for a collision.
- CSMA/CD(collision detection) adds a procedure to handle a collision.
- The station monitors the medium to see if transmission was successful.
- If there was a collision, the station back off a waiting time. Then, send again.
- The waiting time is between 0 and (2^N -1) x propagation_time (N is the number of attempt)

exponential back off

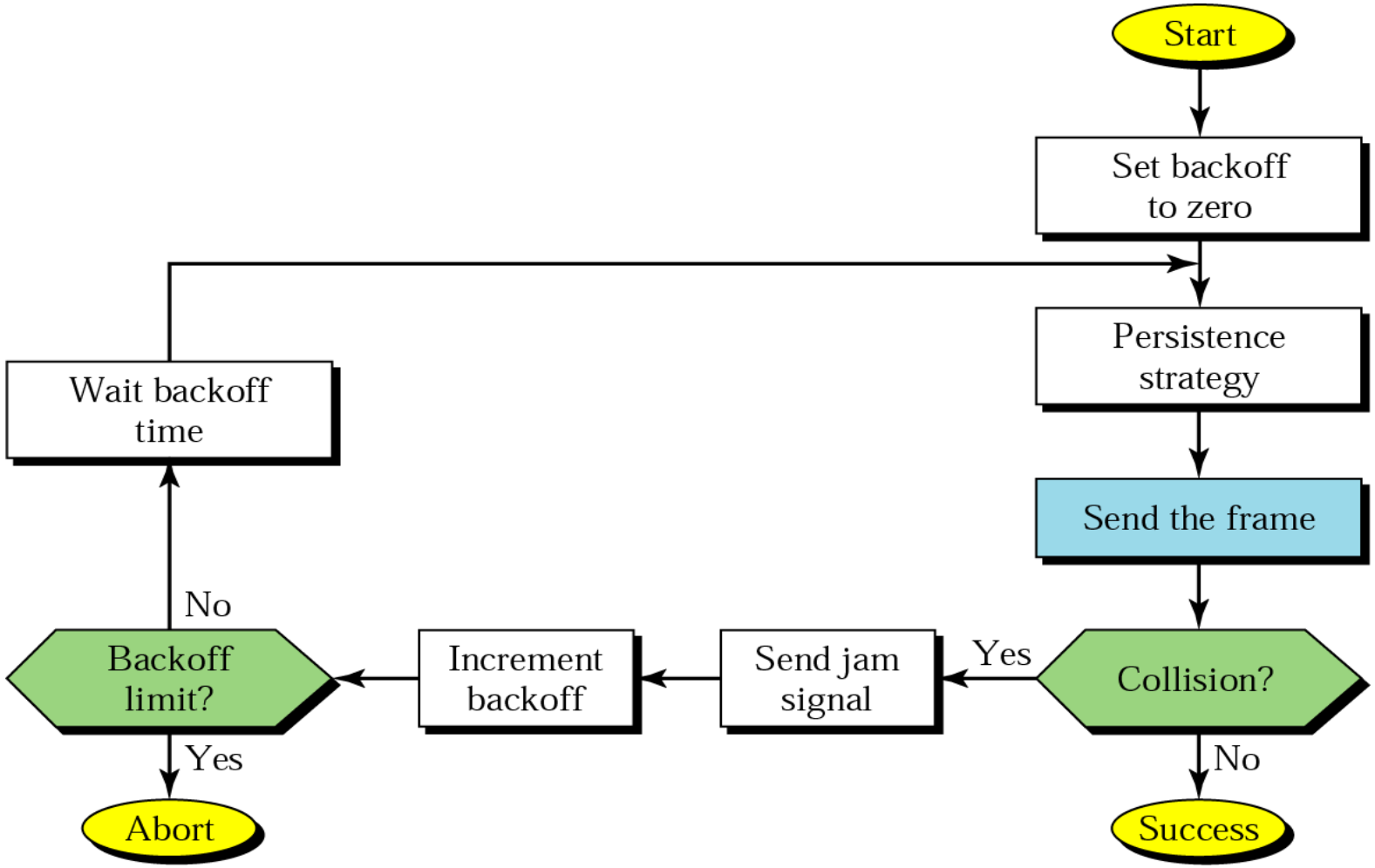
Collision of the First Bit of CSMA/CD



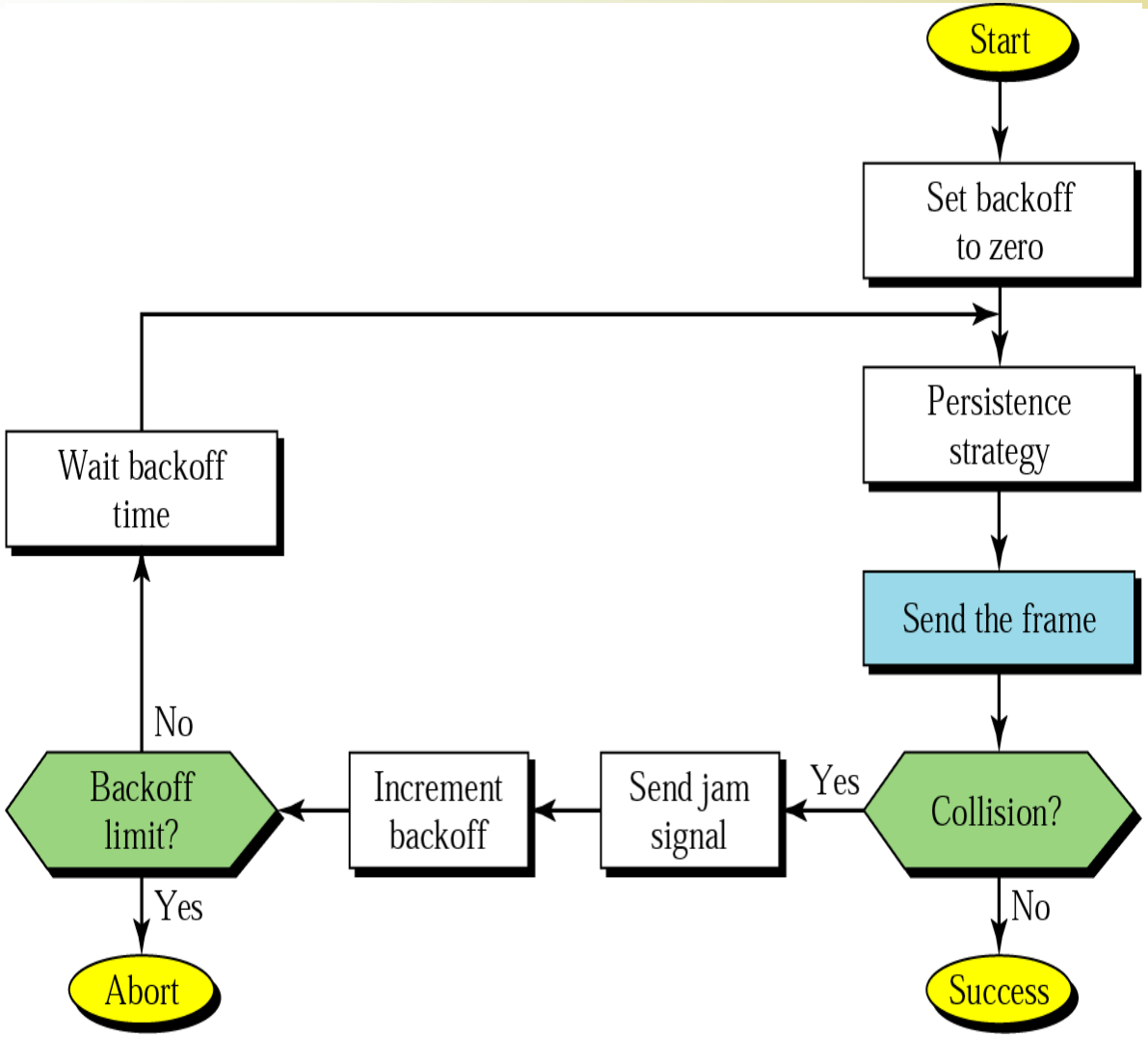
Collision and Abortion in CSMA/CD

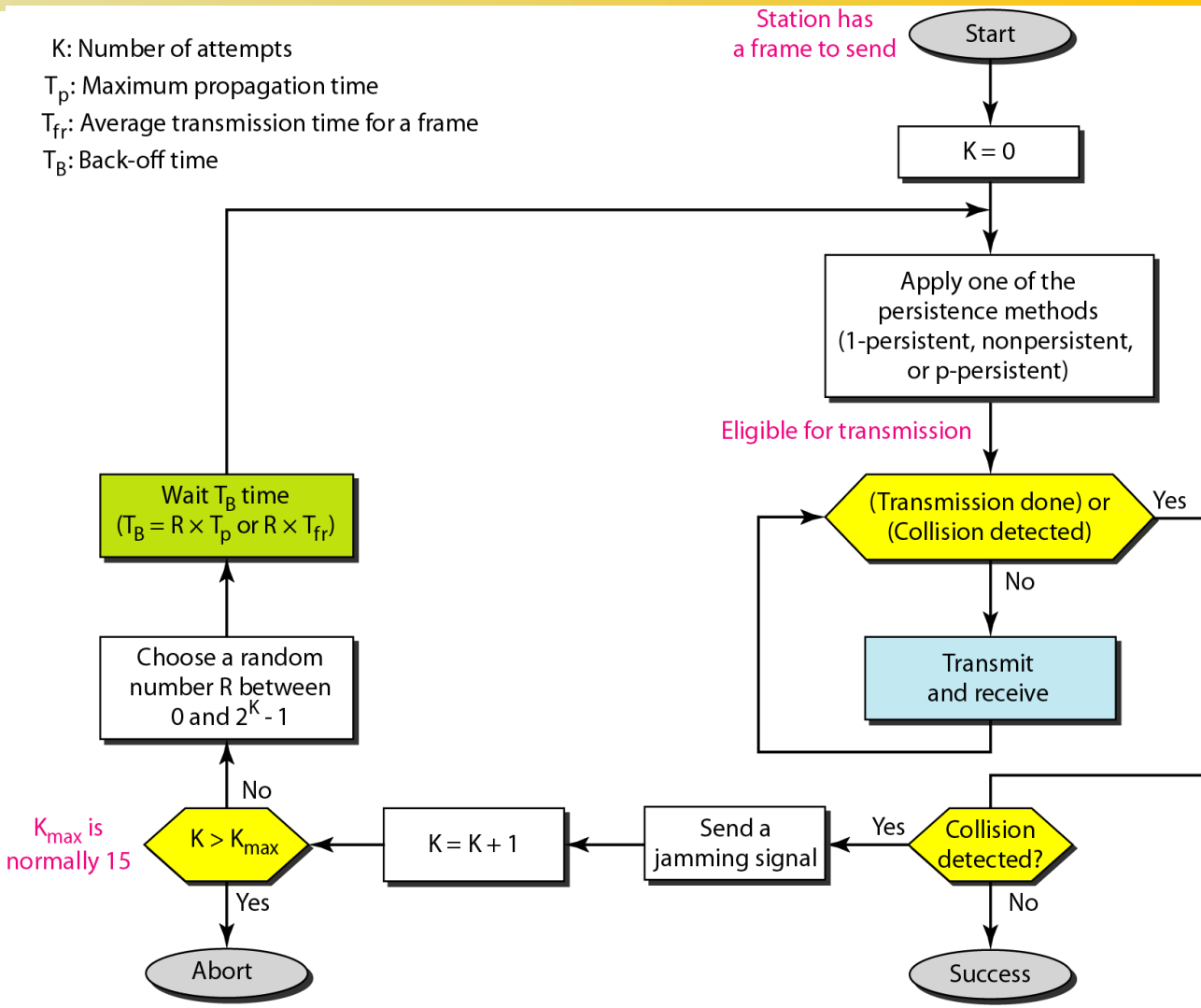


Procedure: Used in Traditional Ethernet

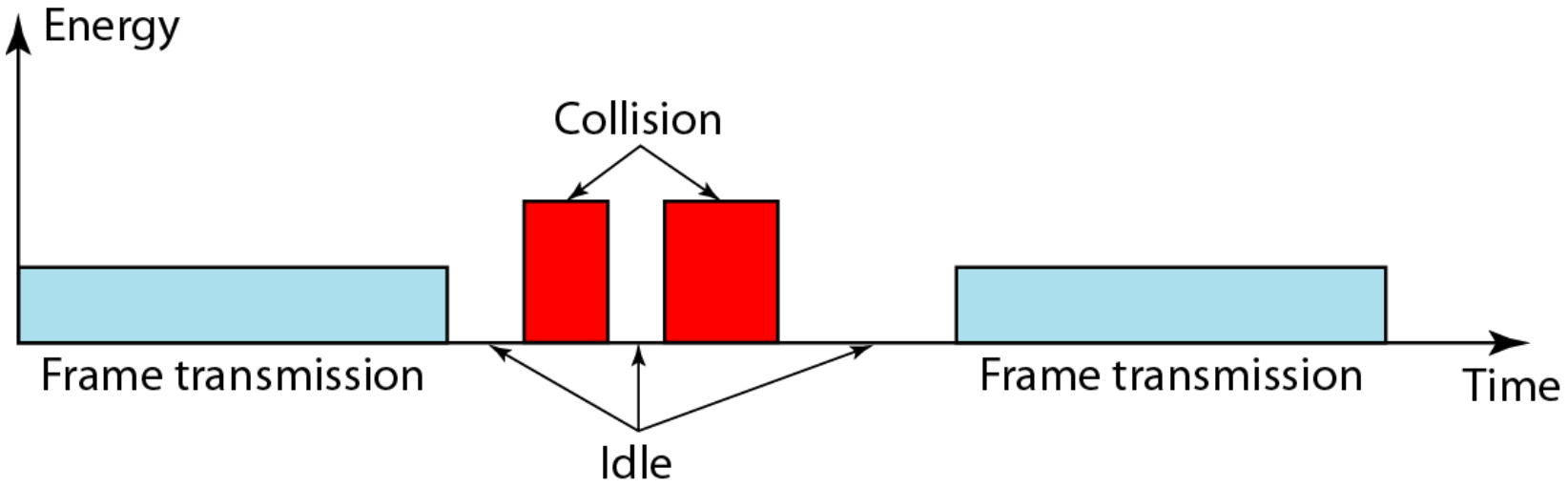


Procedure: Used in Traditional Ethernet





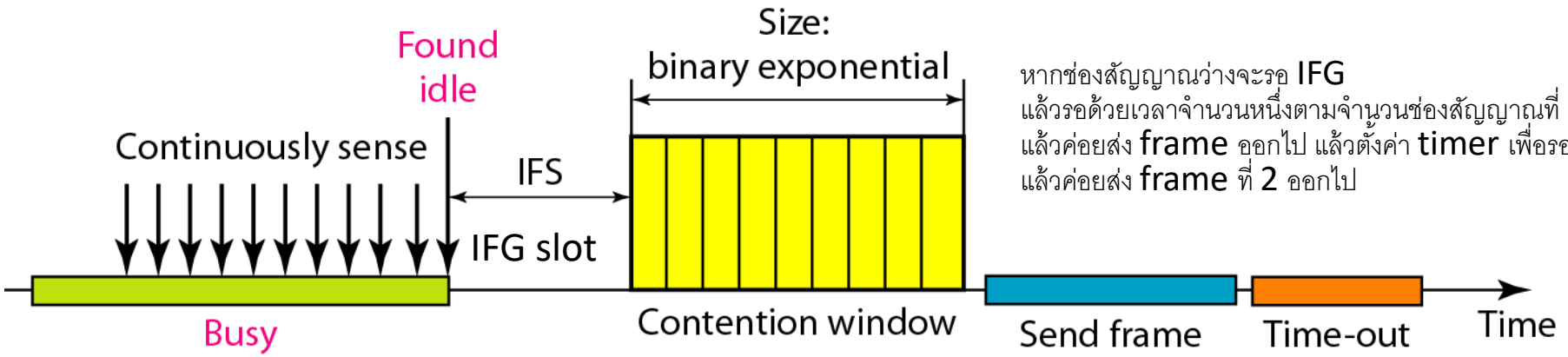
Energy Level during: normal Transmission – idleness - Collision



CSMA/CA

- CSMA/CA differs from the CSMA/CD in that there is no collision detection!
- The station uses one of the persistence strategies
- After it finds the line idle, it waits an *IFG* (_interframe_gap) amount of time
- Then wait another random of time
- Send the frame, and wait for the ACK
- CSMA/CA is used in wireless LAN

IFG & Contention Window



หากช่องสัญญาณว่างจะรอ IFG
แล้วรอด้วยเวลาจำนวนหนึ่งตามจำนวนช่องสัญญาณที่ random
แล้วค่อยส่ง frame ออกไป แล้วตั้งค่า timer เพื่อรอ Ack จากผู้รับ
แล้วค่อยส่ง frame ที่ 2 ออกไป

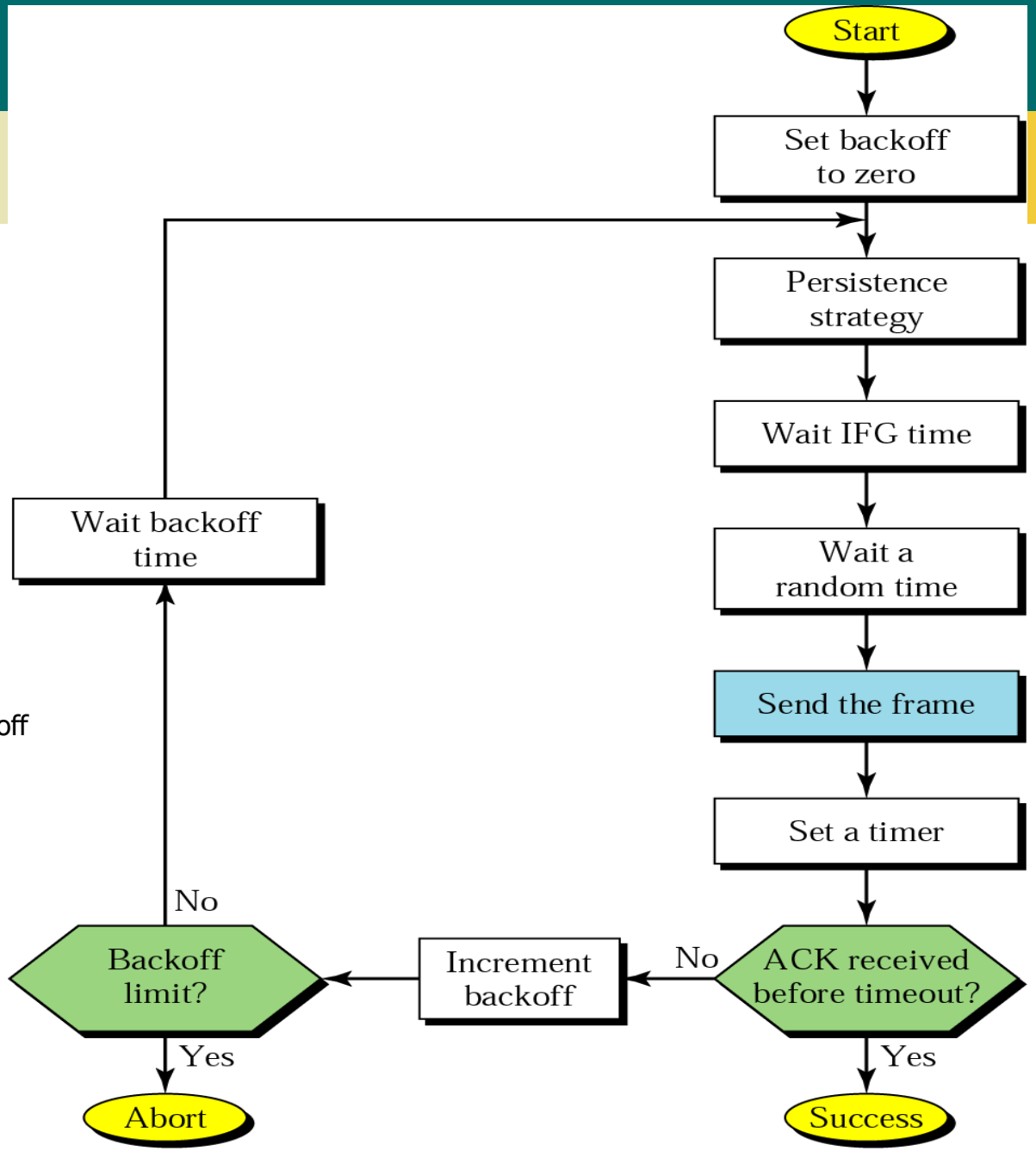
- In CSMA/CA, the **IFS** can also be used to define the priority of a station or a frame. (IFG ที่มีค่าสูงกว่า จะมี priority ต่ำกว่า)
- In CSMA/CA, if the station finds the channel busy, it does not restart the timer of the contention window; it stops the timer and restarts it when the channel becomes idle.
- The contention window is an amount of time divided into slots. A station that is ready to send chooses a random number of slots as its wait time. The number of slots in the window changes according to the binary exponential **backoff strategy**. This means that it is set to one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time.

https://www.youtube.com/watch?v=F9iaRgROFfg

CSMA/CA

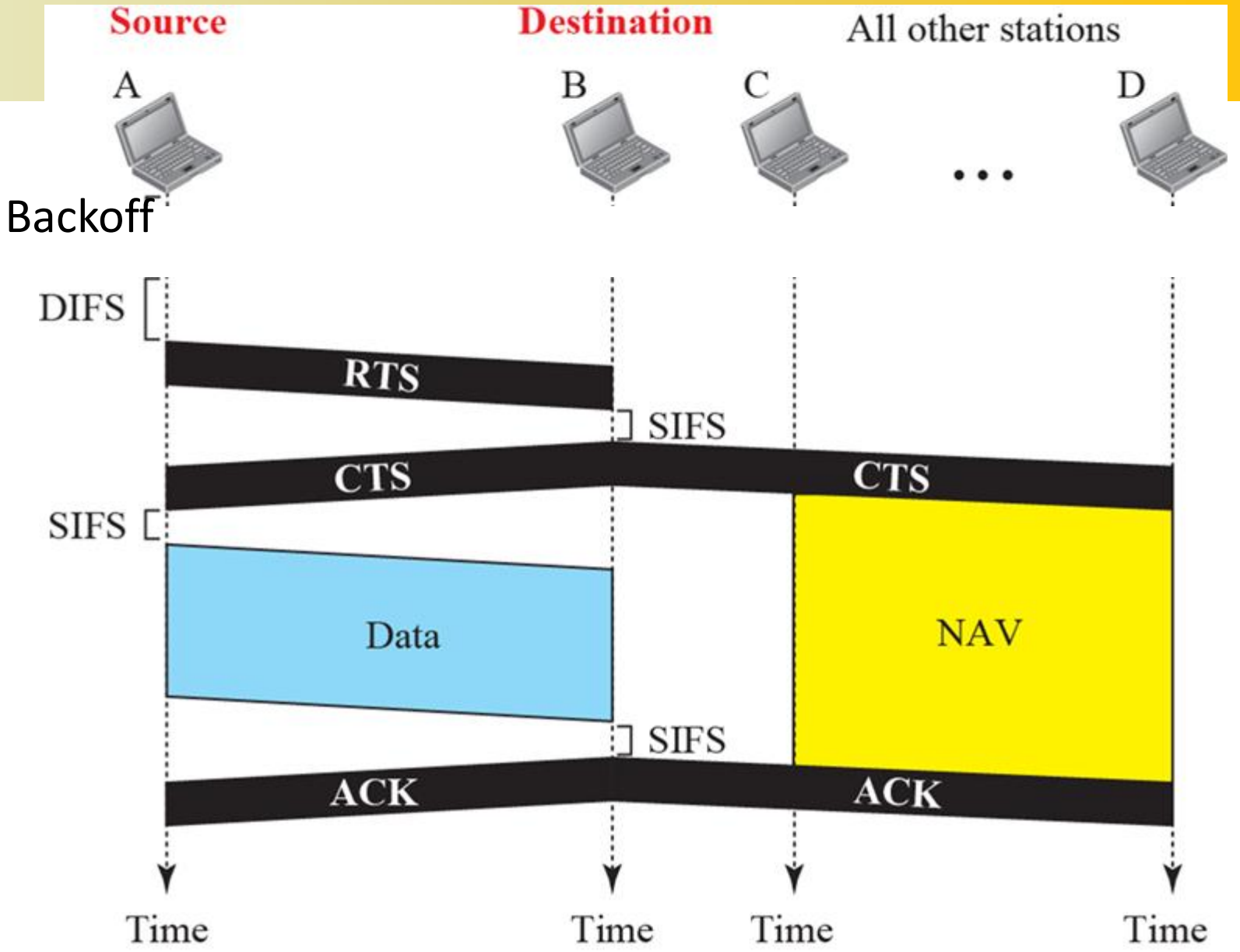
Procedure

หากช่องสัญญาณว่าง
Persistent Strategy: ถ้าใช้ p persistent แล้ว R <=p
Wait IFG time: ช่องสัญญาณรอ IFG
แล้วรอด้วยเวลาจำนวนหนึ่งตามจำนวนช่องสัญญาณที่ random
แล้วค่อยส่ง frame ออกไป แล้วตั้งค่า timer เพื่อรอ Ack จากผู้รับ
หากไม่ได้รับ Ack ในเวลา timer ต้องเพิ่ม backoff แล้วรอด้วยเวลา backoff
หรือถ้าได้รับ Ack สามารถส่ง frame ที่ 2 ต่อไปได้



Procedure with RTS and NAV

- DCF: Distributed Coordination Function
- DIFS: DCF Interframe Spacing
- SIFS: short inter-frame space
- NAV: network allocation vector
- RTS: Request to Send
- CTS: Clear to Send



CSMA/CA

Procedure with RTS

1. Before sending a frame, the source station senses the medium by checking the energy level at the carrier frequency.
   - The channel uses a persistence strategy with backoff until the channel is idle.
   - After the station is found to be idle, the station waits for a period of time called the **DCF interframe space (DIFS)**; then the station sends a control frame called the **request to send (RTS)**

2. After receiving the RTS and waiting a period of time called the ***short interframe space (SIFS),*** the destination station sends a control frame, called the ***clear to send (CTS)***, to the source station.

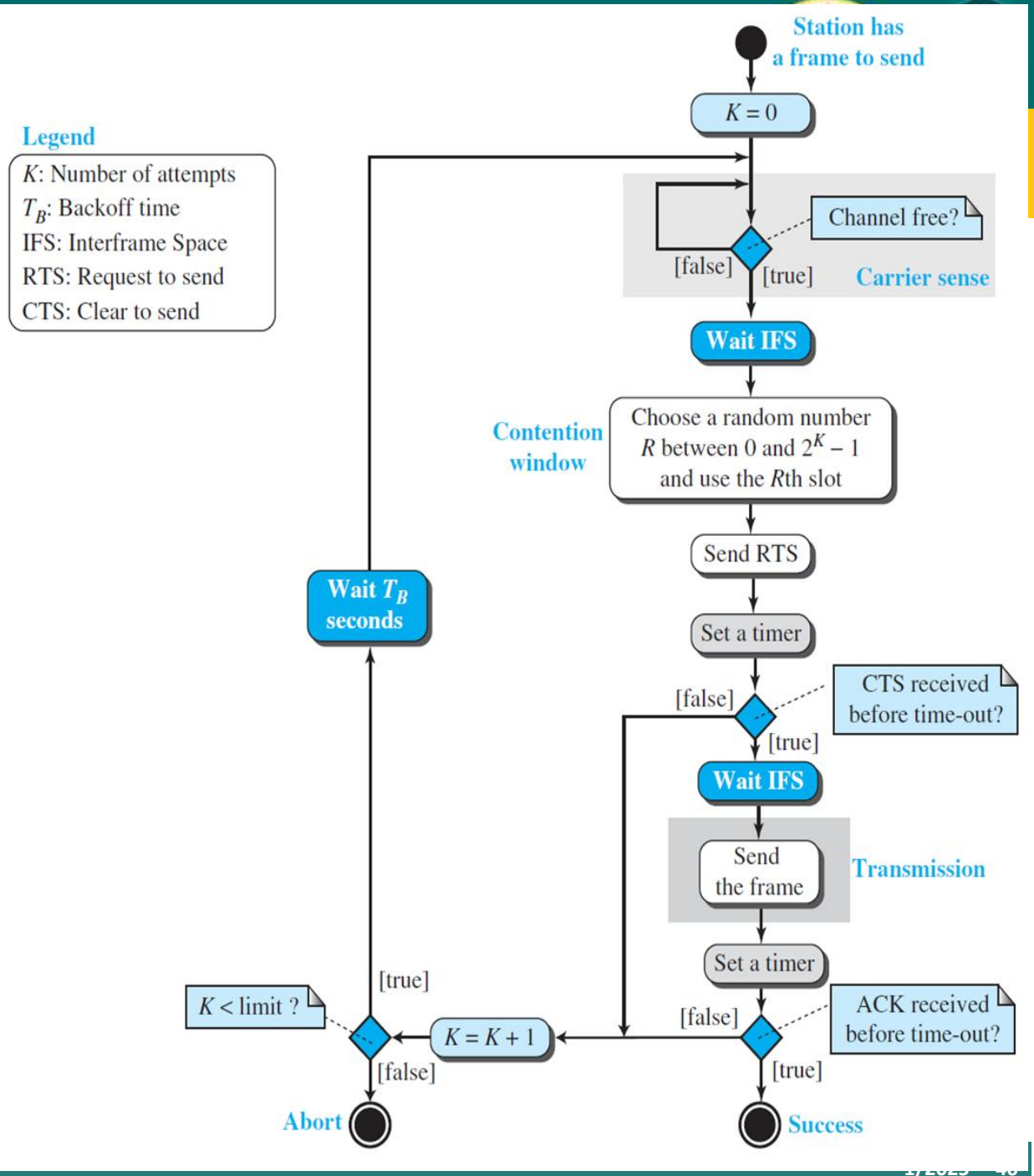
   This control frame indicates that the destination station is ready to receive data.

3. The source station **sends** data **after** waiting an amount of time equal to **SIFS**.

4. The destination station, after waiting an amount of time equal to SIFS, sends an acknowledgment to show that the frame has been received.

   Acknowledgment is needed in this protocol because the station does not have any means to check for the successful arrival of its data at the destination.

   On the other hand, the lack of collision in CSMA/CD is a kind of indication to the source that data have arrived.



Procedure with NAV

- How do other stations defer sending their data if one station acquires access? In other words, how is the collision avoidance aspect of this protocol accomplished? The key is a feature called NAV.

- When a station sends an RTS frame, it includes the duration of time that it needs to occupy the channel.
  - The stations that are affected by this transmission create a timer called a **network allocation vector (NAV)** that shows how much time must pass before these stations are allowed to check the channel for idleness.
  - Each time a station accesses the system and sends an RTS frame, other stations start their NAV.
  - In other words, each station, before sensing the physical medium to see if it is idle, first checks its NAV to see if it has expired. Figure in the previous diagram shows the idea of NAV.

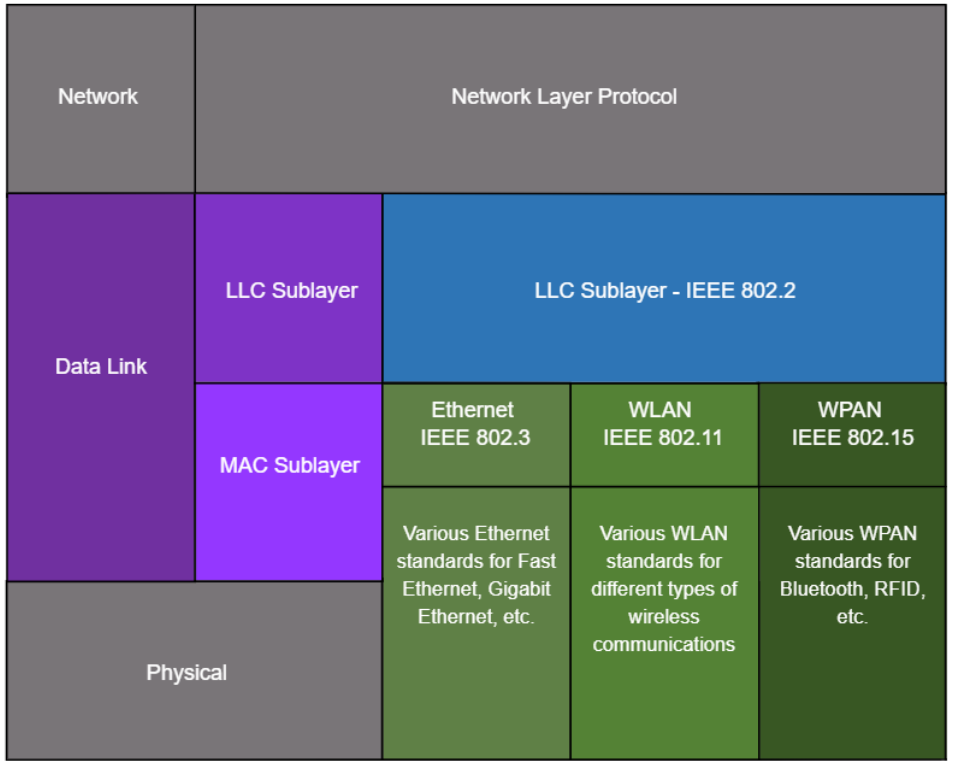
Collision during Handshaking

- What happens if there is a collision during the time when RTS or CTS control frames are in transition, often called the handshaking period?

- Two or more stations may try to send RTS frames at the same time. These control frames may collide. However, because there is no mechanism for collision detection, the sender assumes there has been a collision if it has not received a CTS frame from the receiver. The backoff strategy is employed, and the sender tries again.

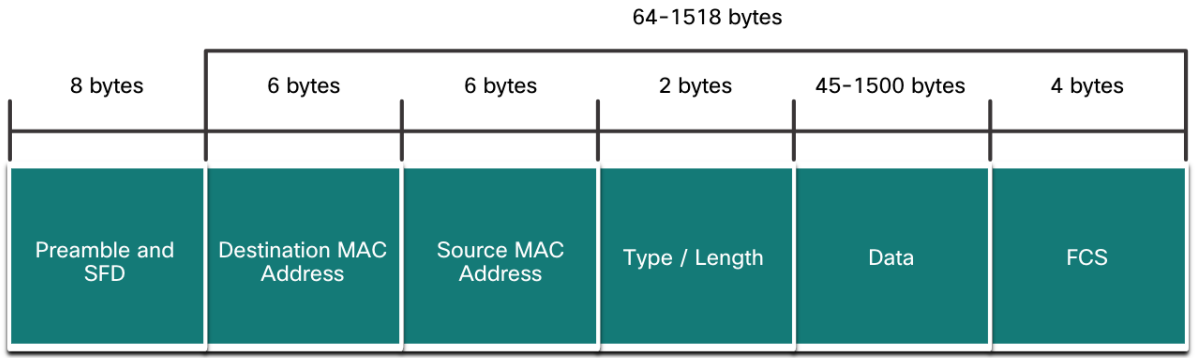
- Hidden Node Problem: This can help solve the hidden communication problem. (A talk to B, but C and D knows about the communication in session from the CTS that B sends to A.

IEEE 802 LAN/MAN



- IEEE 802 LAN/MAN standards are specific to the type of network (Ethernet, WLAN, WPAN, etc).
- The Data Link Layer consists of two sublayers. **Logical Link Control (LLC)** and **Media Access Control (MAC).**
  - The LLC sublayer communicates between the networking software at the upper layers and the device hardware at the lower layers.
  - The MAC sublayer is responsible for data encapsulation and media access control.
- IEEE 802.3 data encapsulation includes the following:
  - ✓ **Ethernet frame** - This is the internal structure of the Ethernet frame.
  - ✓ **Ethernet Addressing** - The Ethernet frame includes both a source and destination MAC address to deliver the Ethernet frame from Ethernet NIC to Ethernet NIC on the same LAN.
  - ✓ **Ethernet Error detection** - The Ethernet frame includes a frame check sequence (FCS) trailer used for error detection.

Frame Fields



- The minimum Ethernet frame size is 64 bytes and the maximum is 1518 bytes.
- The preamble field is not included when describing the size of the frame.
  - Any frame less than 64 bytes in length is considered a “collision fragment” or “runt frame” and is automatically discarded.
  - Frames with more than 1500 bytes of data are considered “jumbo” or “baby giant frames”.
- If the size of a transmitted frame is less than the minimum, or greater than the maximum, the receiving device drops the frame. Dropped frames are likely to be the result of collisions or other unwanted signals. They are considered invalid.
- Jumbo frames are usually supported by most Fast Ethernet and Gigabit Ethernet switches and NICs.

The Minimum Size of the frame 10 Mbps is 64 Bytes

A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal) is 25.6 µs, what is the minimum size of the frame?

Solution

- The minimum frame transmission time is $T_{fr} = 2 \times T_p = 51.2$ µs. This means, in the worst case, a station needs to transmit for a period of 51.2 µs to detect the collision.

- The minimum size of the frame is 10 Mbps × 51.2 µs = 512 bits or 64 bytes.
- This is actually the minimum size of the frame for Standard Ethernet, as we will see later in the chapter.

IEEE 802.11 vs WiFi

- WiFi = Wireless Fidelity is a trademark
- Fidelity = Compatibility between wireless equipment from different manufacturders
- WiFi Alliance is a non-profit organization that does the compatibility testing (WiFi.org)
- 802.11 has many options and it is possible for two equipment based on 802.11 to be incompatible.

Overview: https://en.wikipedia.org/wiki/IEEE_802.11
Working group: http://www.ieee802.org/11/

Controlled Access

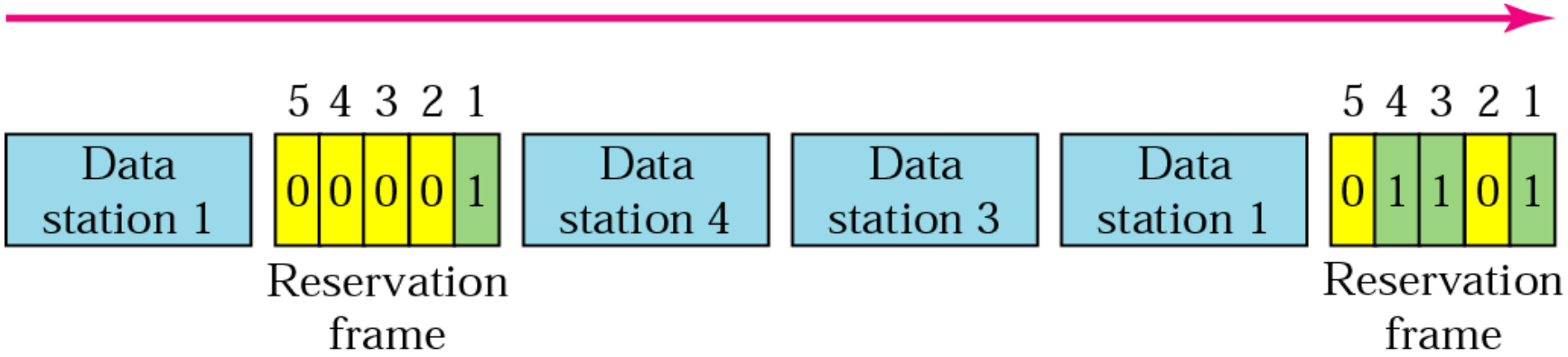
Controlled Access

- In controlled access, the station consult one another to find which station has the right to send
- A station cannot send unless it has been authorized by other stations
  - ✓ Reservation
  - ✓ Polling (Round-Robin based)
  - ✓ Token Passing (Round-Robin based)

Stations are allowed "turns" to transmit for a certain maximum period

Reservation

- A station needs to make a reservation before sending data
- In each interval, a reservation frame precedes the data frames sent in that interval
  - ✓ If there are *N* stations in the system, there are *N* reservation minislots in the reservation frame
- When a station needs to send a data frame, it makes a reservation in its own minislot
- **Reservation Access Method**: The stations that have made reservations can send their data frame after the reservation frame.



Polling: Select and Poll

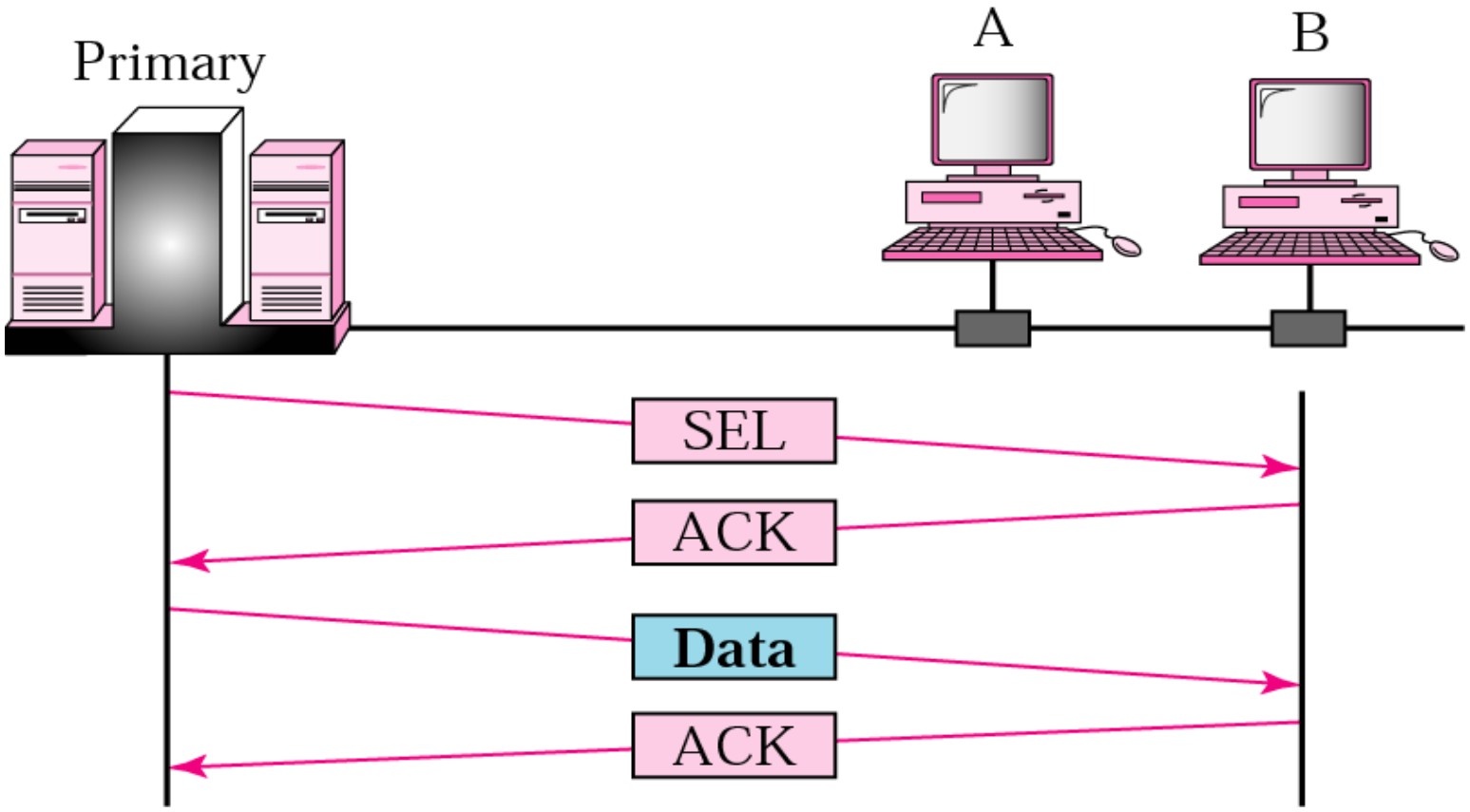
- Polling works with topologies in which one device is designed as a primary station (master) and the other devices are secondary stations (slaves)

- All data exchanges must be made through the primary device
  - ✓ The primary device controls the links and determine which device is allowed to use the channel at a given time
  - ✓ The secondary devices follow its instructions

Controlled Access

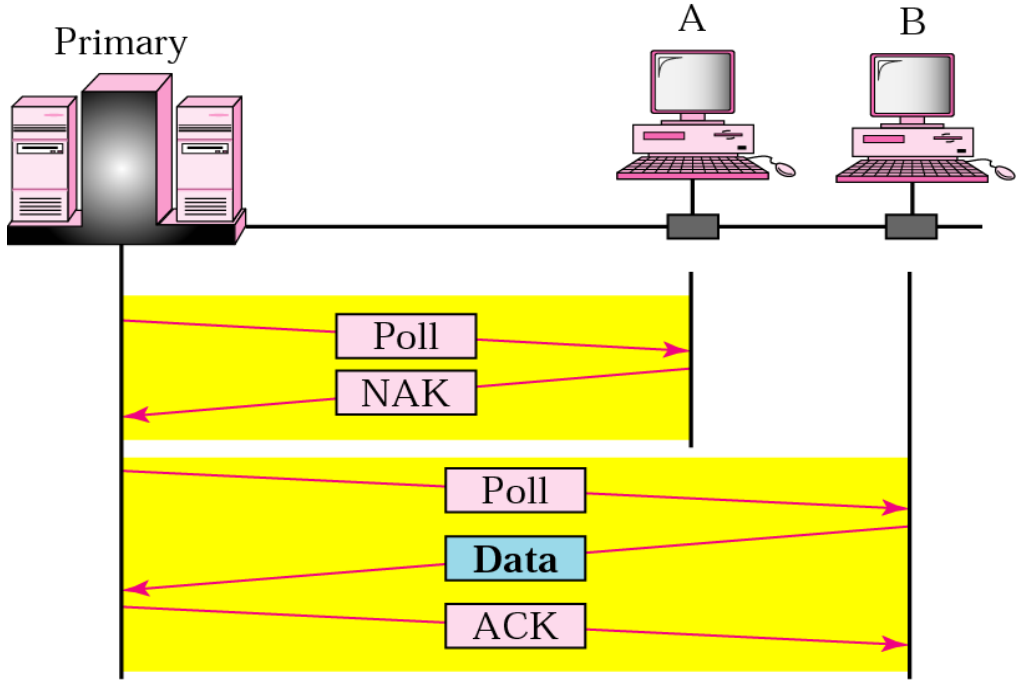
Polling: Select

- Whenever the primary device has something to send, it alerts the secondary to the upcoming transmission and wait for an ACK of the ready status from the secondary.



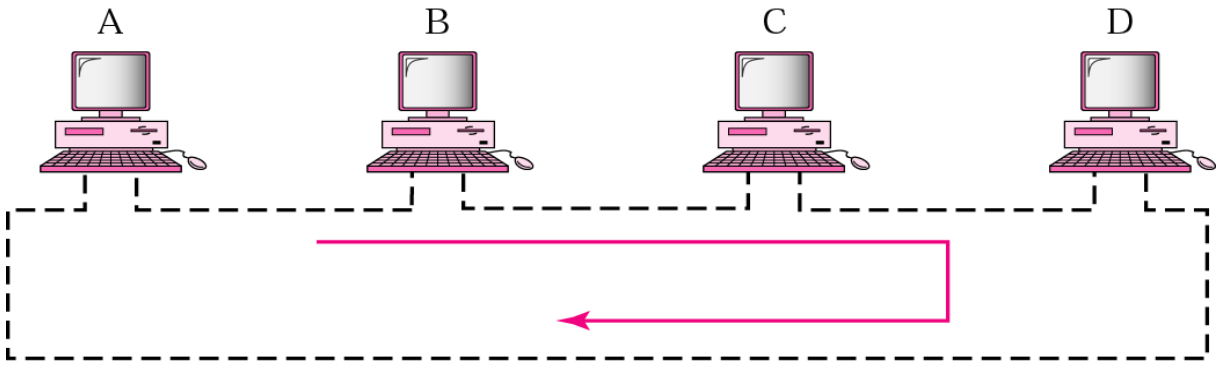
Polling: Poll

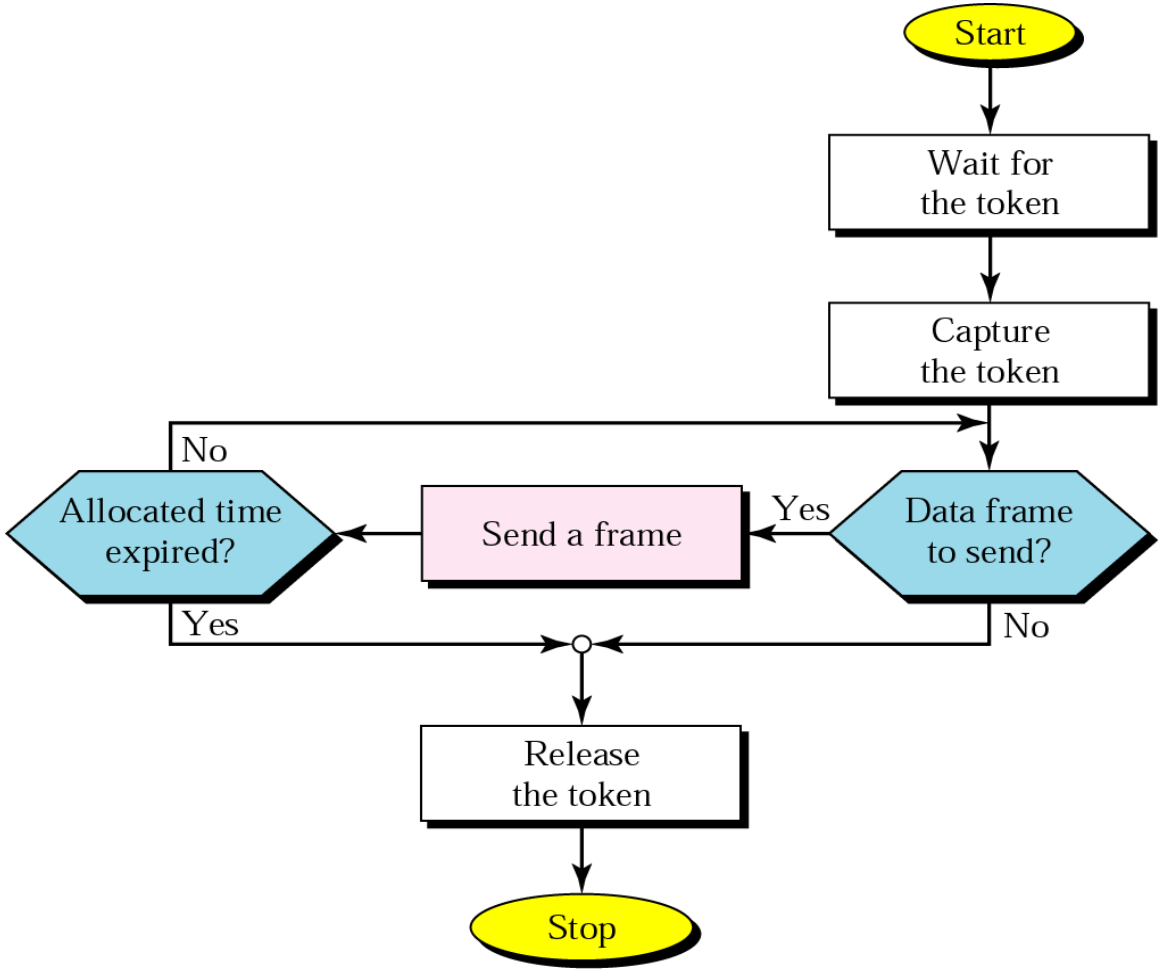
- Used by the primary device to solicit transmissions from the secondary device
- When the primary is ready to receive data, it asks each device whether it has anything to send
- If the response is NAK, the primary then polls the next secondary



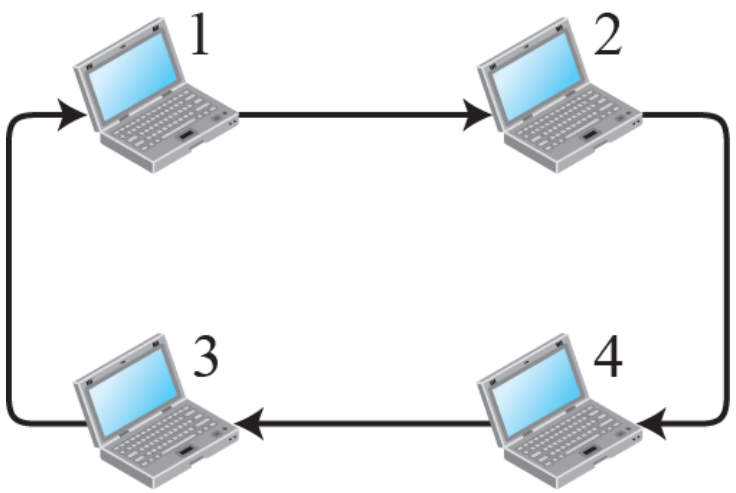
Token Passing

- Stations are arranged round a **ring**
  - ✓ Successor
  - ✓ Predecessor
- A station is authorized to send data when it receives a special frame called a **token**

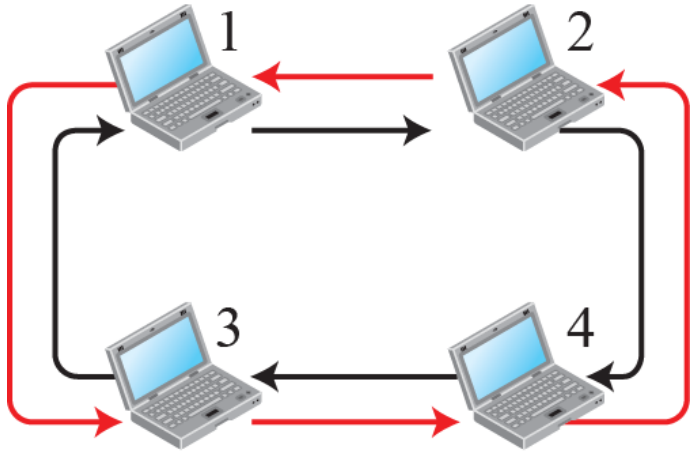




Physical and Logical Ring Topology in Token Passing



a. Physical ring

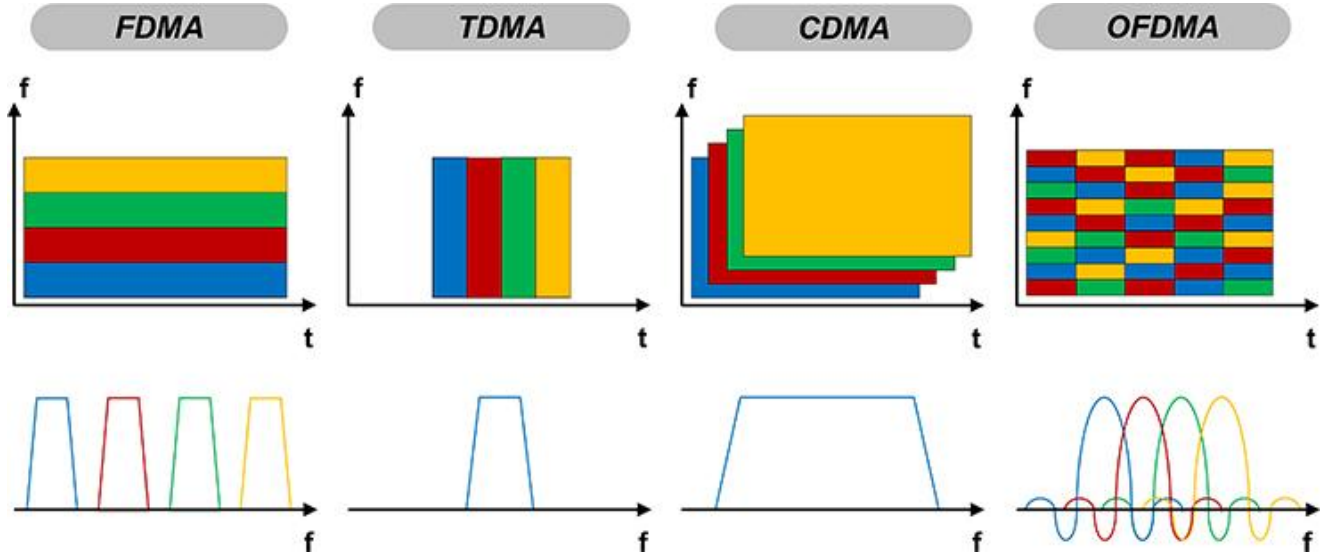


b. Dual ring

Channelization

Channelization

- Channelization is a multiple-access method in which the available bandwidth of a link is shared between different stations
  - FDMA (Frequency-Division Multiple Access)
  - TDMA (Time-Division Multiple Access)
  - CDMA (Code-Division Multiple Access)
  - OFDMA (Orthogonal Frequency Division Multiple Access)



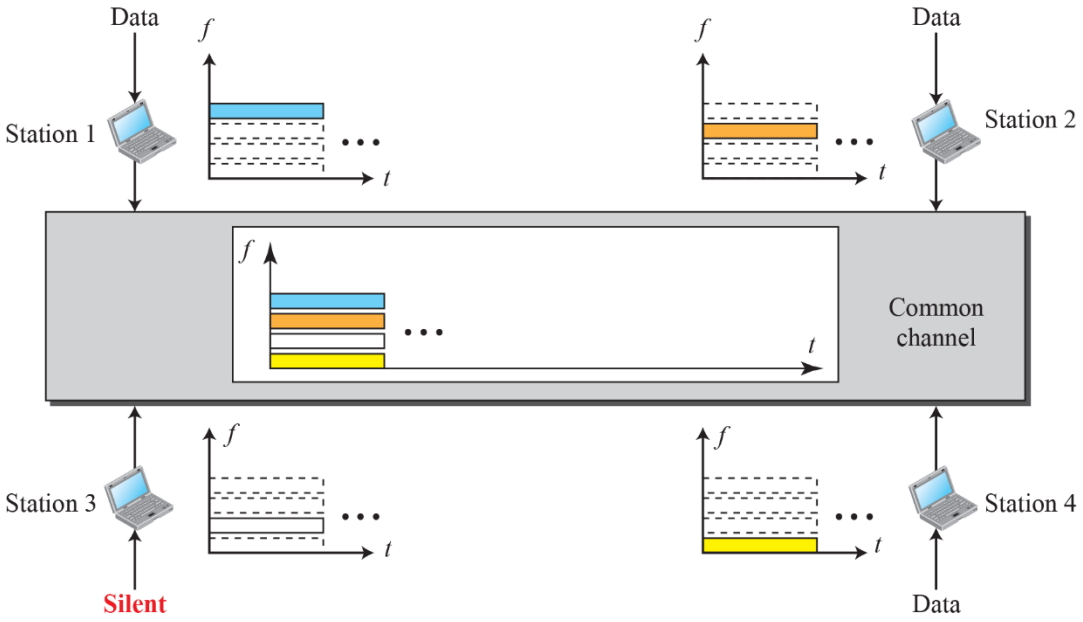
Channelization

COMPARISON OF 1G TO 5G TECHNOLOGIES

Technology	1G	2G/2.5G	3G	4G	5G
Deployment	1970/1984	1980/1999	1990/2002	2000/2010	2014/2015
Bandwidth	2kbps	14-64kbps	2mbps	200mbps	>1gbps
Technology	Analog cellular	Digital cellular	Broadbandwidth/ cdma/ip technology	Unified ip & seamless combo of LAN/WAN/WLAN/PAN	4G+WWWW
Service	Mobile telephony	Digital voice,short messaging	Integrated high quality audio, video & data	Dynamic information access, variable devices	Dynamic information access, variable devices with AI capabilities
Multiplexing	FDMA	TDMA/CDMA	CDMA	CDMA	CDMA
Switching	Circuit	Circuit/circuit for access network& air interface	Packet except for air interface	All packet	All packet
Core network	PSTN	PSTN	Packet network	Internet	Internet
Handoff	Horizontal	Horizontal	Horizontal	Horizontal&Vertical	Horizontal&Vertical

FDMA (Frequency-Division Multiple Access)

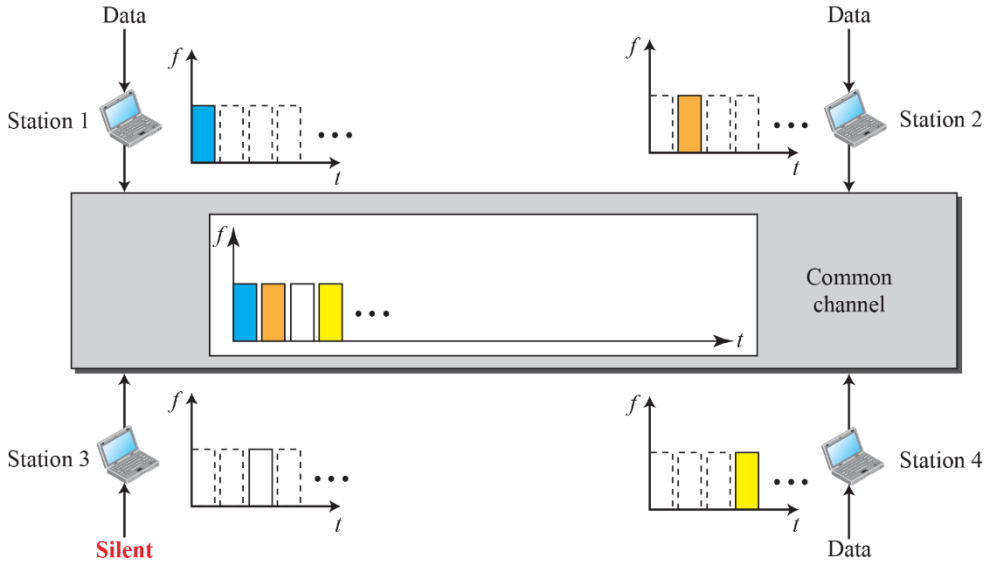
- The available bandwidth is shared by all stations
- FDMA is a data link layer protocol that uses FDM at the physical layer (Lecture 6)
- Used in cellular phone and satellite networks



In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.

TDMA (Time-Division Multiple Access)

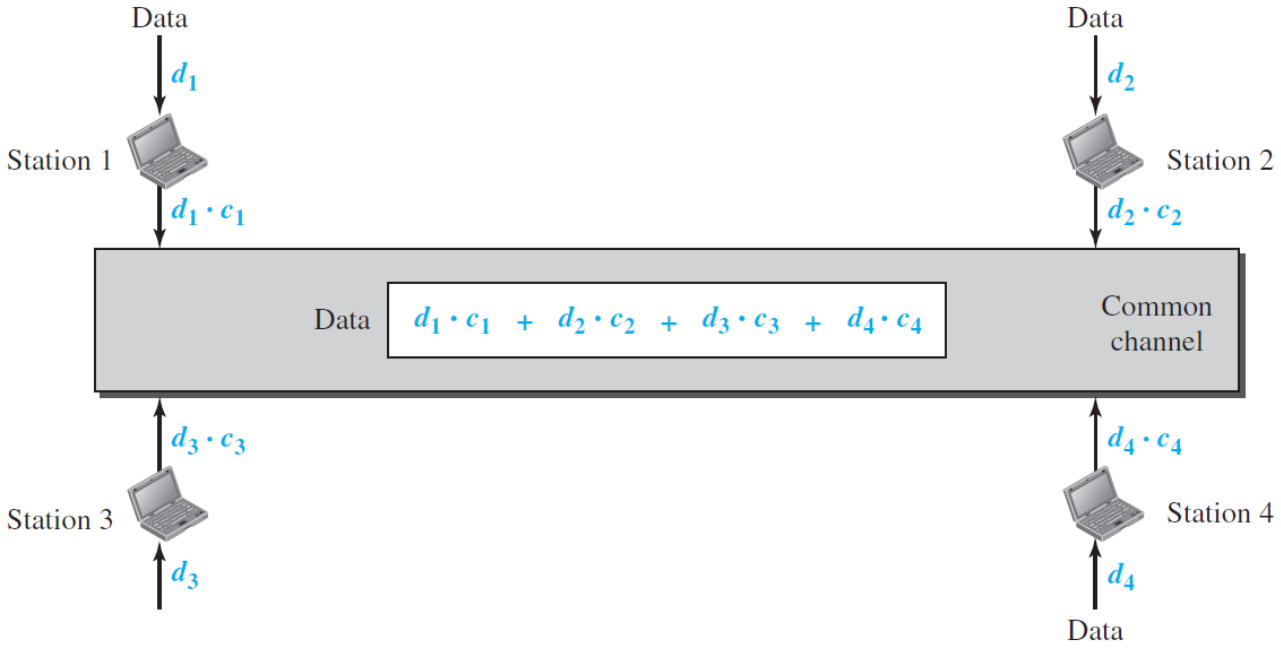
- The entire bandwidth is just one channel
- The stations share the capacity of the channel in time
- TDMA is a data link layer protocol that uses TDM at the physical layer



In TDMA, the bandwidth is just one channel that is timeshared between different stations.

CDMA (Code-Division Multiple Access)

- Differs from FDMA because only one channel occupies the entire bandwidth of the link
- Differs from TDMA because all stations can send data simultaneously; there is no timesharing
- Used in the cellular telephone network



In CDMA, one channel carries all transmissions simultaneously.

CDMA (Code-Division Multiple Access)

- CDMA is based on *coding theory*
- Each station is assigned a code, which is a sequence of numbers called chips
- Suppose we have four stations; each has a sequence of chips which we designate as A, B, C, D

Orthogonal Chip sequences

A	B	C	D
+1, +1, +1, +1	+1, −1, +1, −1	+1, +1, −1, −1	+1, −1, −1, +1

CDMA (Code-Division Multiple Access)

- To generate orthogonal sequences, we use a **Walsh table**,
  a two-dimensional *table* with an equal number of rows and column
- Each row is a sequence of chips
- W_n for a *n*-chip sequence has *n* row and *n* column

Orthogonal Chip sequences

A	B	C	D
+1, +1, +1, +1	+1, −1, +1, −1	+1, +1, −1, −1	+1, −1, −1, +1

CDMA (Code-Division Multiple Access)

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

Wn with the overhead bar stands for the complement of Wn

CDMA (Code-Division Multiple Access): Sequence Generation

$$W_1 = \begin{bmatrix} +\mathbf{1} \end{bmatrix}$$

$$W_2 = \begin{bmatrix} W_1 & W_1 \\ W_1 & \overline{W_1} \end{bmatrix} \qquad W_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$W_4 = \begin{bmatrix} W_2 & W_2 \\ W_2 & \overline{W_2} \end{bmatrix} \qquad W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

Wn with the overhead bar stands for the complement of Wn

CDMA (Code-Division Multiple Access): Encoding Rules

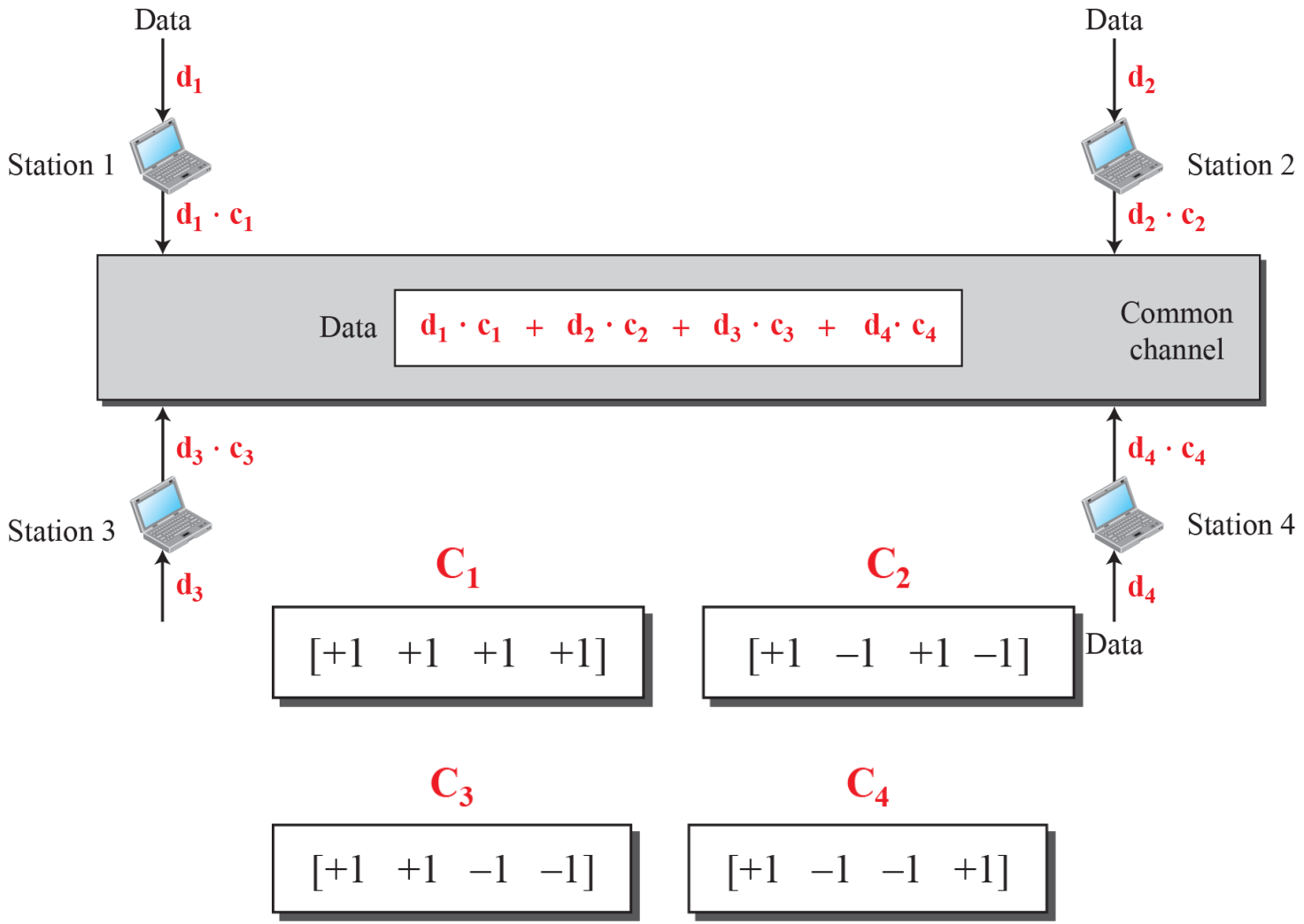
Rules for encoding

- ✓ If a station needs to send a 0 bit, sends a –1
- ✓ if it needs to send a 1 bit, it sends a +1
- ✓ When a station is idle, it sends no signal (0)

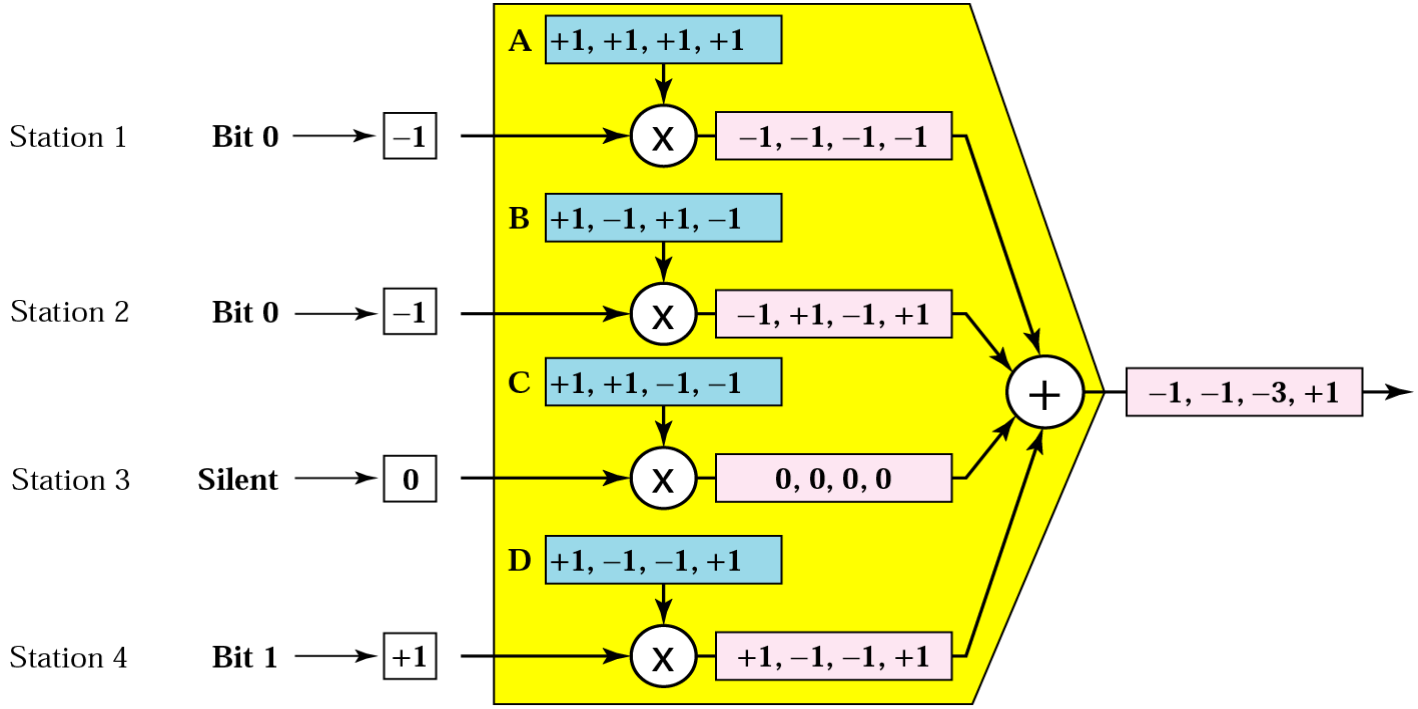
Data bit 0 → –1	Data bit 1 → +1	Silence → 0

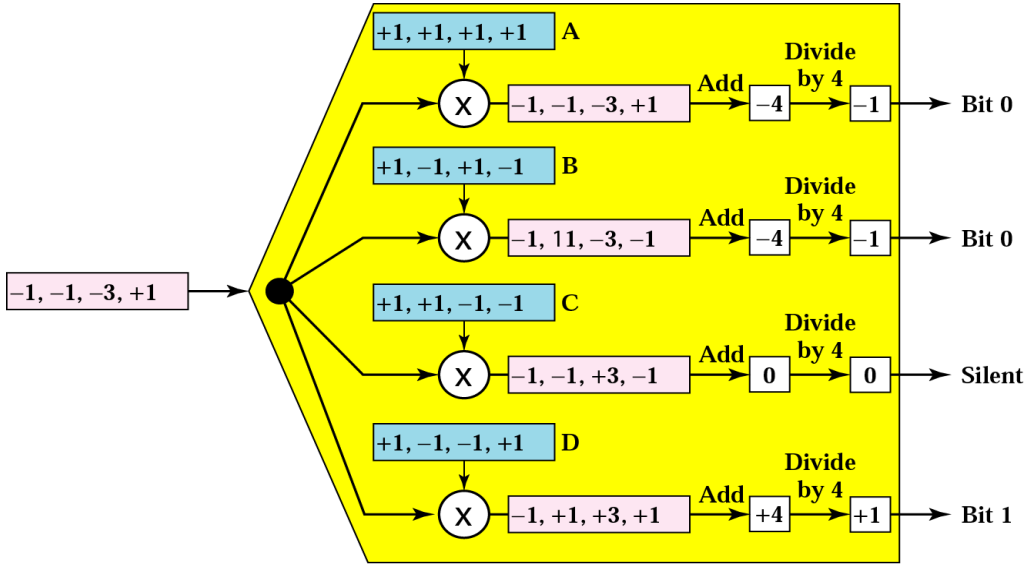
CDMA (Code-Division Multiple Access): Example

$$W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$



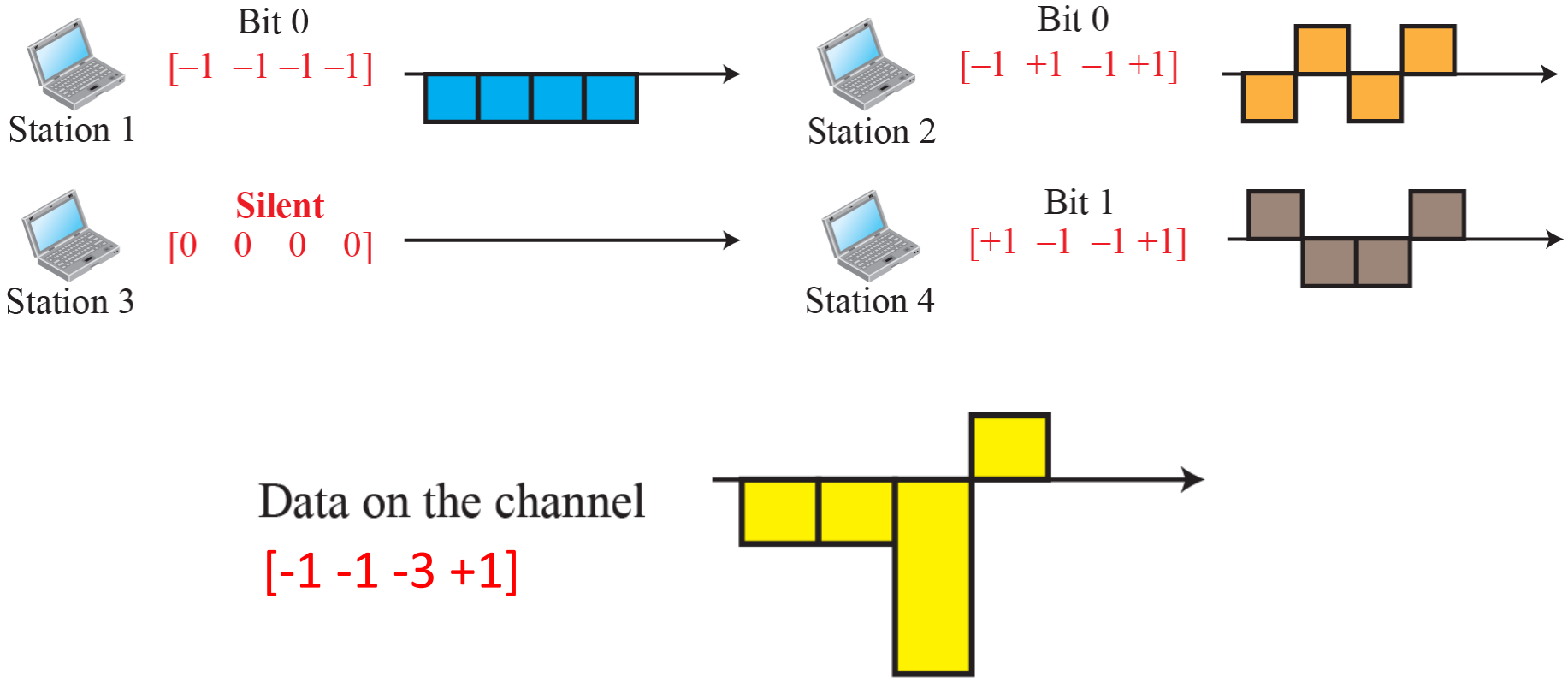
CDMA (Code-Division Multiple Access): Multiplexer



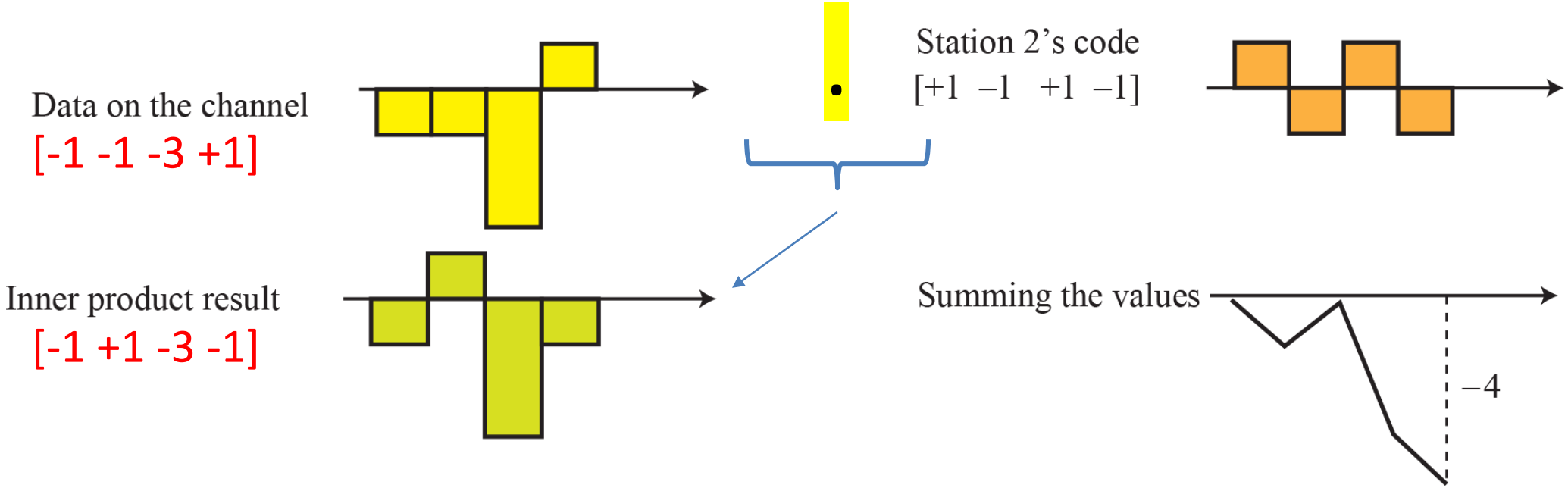


CDMA

- Digital signal created by four stations in CDMA



- Decoding of the composite signal in CDMA



–4 → –4/4 → –1 → Bit 0

Properties of Orthogonal Sequences

- If we multiply a sequence by –1, every element in the sequence is complemented
  - ✓ We can that when a station is sending –1 (bit 0), it is sending its complement

- If we multiply two sequences, element by element, and add the result, we get a number called the **inner product**
  - ✓ If the two sequences are the same, we get **N**, where N is the number of sequences
  - ✓ if they are different, we get **0**

- The inner product of a sequence by its complement is **–N**
  - ✓ A · (-A) is -N

Example 1

Check to see if the second property about orthogonal codes holds for our CDMA example.

The inner product of each code by itself is N. This is shown for code C; you can prove for yourself that it holds true for the other codes.

C . C = [+1, +1, -1, -1] . [+1, +1, -1, -1] = 1 + 1 + 1 + 1 = 4

If two sequences are different, the inner product is 0.

B . C = [+1, -1, +1, -1] . [+1, +1, -1, -1] = 1 - 1 - 1 + 1 = 0

Example 2

Check to see if the second and third property about orthogonal codes holds for our CDMA example.

The inner product of each code by its complement is -N.
This is shown for code C; you can prove for yourself that it holds true for the other codes.

C . (-C) = [+1, +1, -1, -1] . [-1, -1, +1, +1] = - 1 - 1 - 1 - 1 = -4

The inner product of a code with the complement of another code is 0.

B . (-C) = [+1, -1, +1, -1] . [-1, -1, +1, +1] = -1 + 1 + 1 - 1 = 0

Example 3

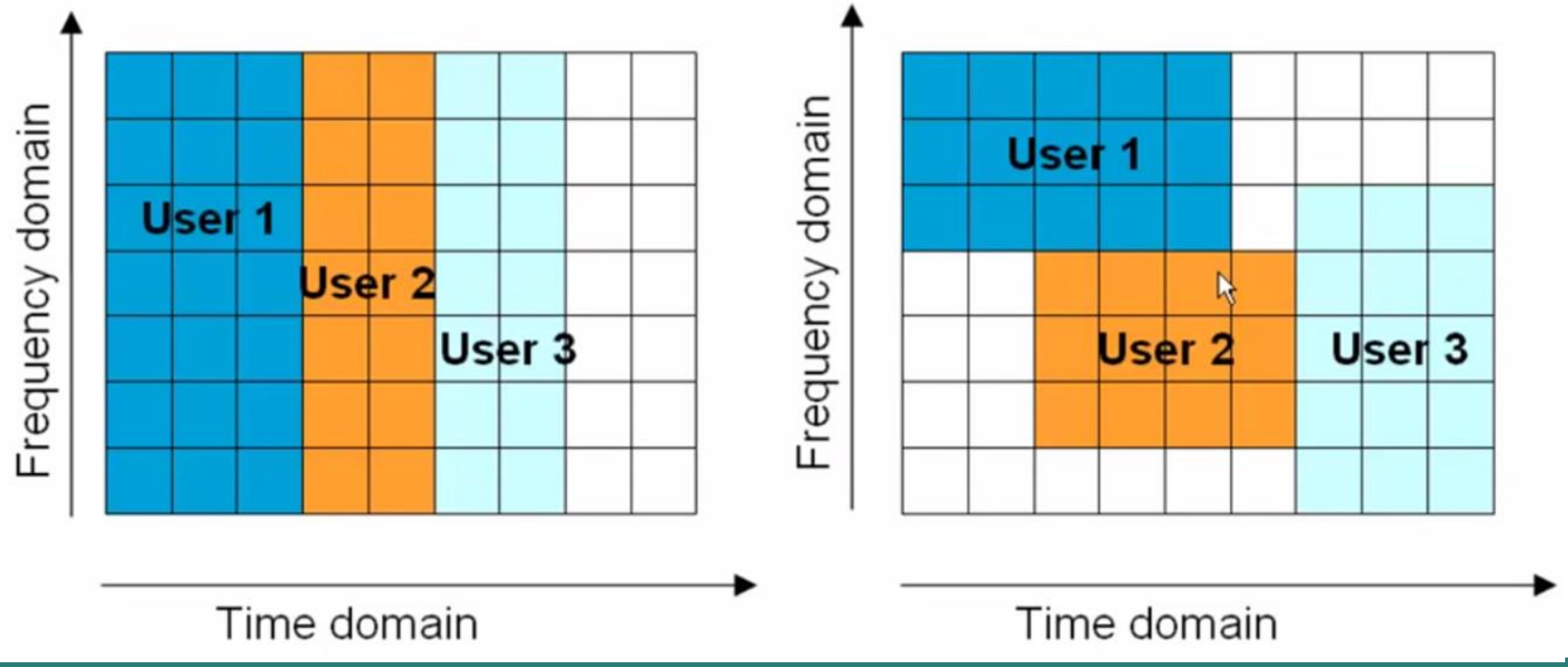
- In the Multiplexer
  - ✓ Each station is sending the appropriate sequence
  - ✓ Station 1 sends –A
  - ✓ Station 2 sends –B
  - ✓ Station 3 is silent
  - ✓ Station 4 sends D
  - ✓ Sum of all sequences S = -A -B + D

- In the Demultiplexer
  - Station 1 finds the inner product of S and A
    - S· A = (- A -B + D) · A
      = -A·A + -B·A + D·A = -4 + 0 + 0 = -4
    - Then the result is divided by 4 → -1 → 0 bit
  - Station 2 finds the inner product of S and B
    - S·B = (-A -B +D) · B
      = -A·B -B·B +D·B = 0 -4 + 0 = -4 → 0 bit
  - Station 3 finds the inner product of S and C
    - S·C = (-A –B + D) ·C = 0 → Silent
  - Station 4 finds the inner product of S and D
    - S·D = (-A –B+D) ·D = 0 + 0 + 4 → 1 bit

OFDM (Orthogonal Frequency Division Multiplexing)

- In **OFDM**, frequency and time can both be splitting to create a grid of communication allowing more senders and receivers (users) in the network and increase bandwidth.

- **OFDM** is used extensively in modern wireless networks **at the physical layers** such as 802.11ac, 802.11ax (Wi-Fi 6), 4G, and 5G networks



- Different users can have different slices of time and different groups of subcarriers
- Subcarriers are allocated in groups:
  - ✓ Called subchannels or resource blocks.
  - ✓ Too much computation to allocate every subcarrier separately